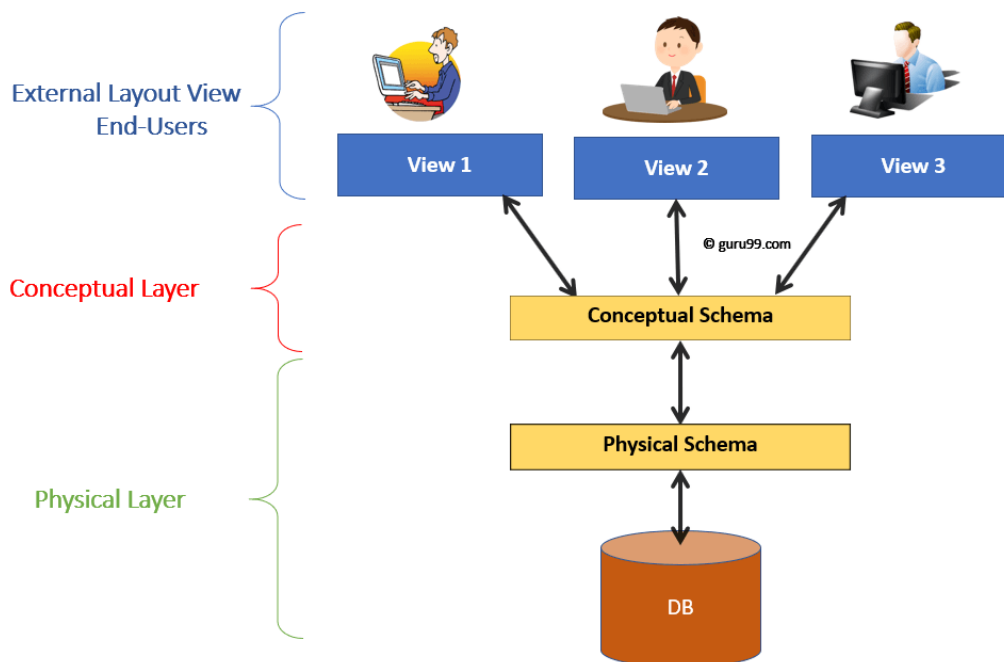


Database System Architecture:

The "database architecture" refers to the structural design and methodology of a database system, which forms the core of a Database Management System (DBMS). This architecture dictates how data is stored, organized, and retrieved, playing a crucial role in the efficiency and effectiveness of data management.

Database architecture describes how a database management system (DBMS) will be integrated with your application. When designing database architecture, you will make decisions that will change how your applications are created.



DBMS architecture refers to the structural design and components that work in harmony to manage and maintain databases effectively. One common approach is the client-server architecture, which separates client and server components to facilitate data handling, application logic, and user interactions.

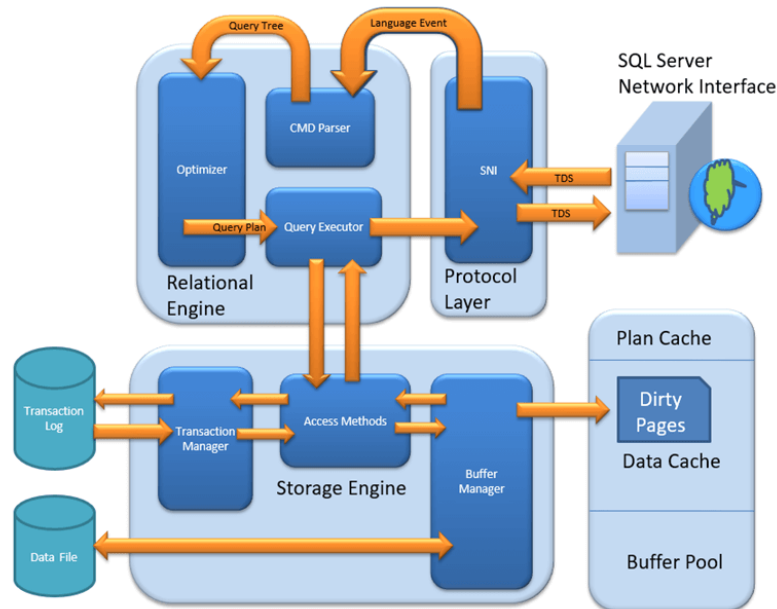
At the core of DBMS architecture are several key components, including the data dictionary, query processor, transaction manager, and storage manager.

The **data dictionary** acts as the central repository for metadata, providing information about the database's structure and organization. The **query processor** is responsible for interpreting and executing user queries, while the **transaction manager** ensures data integrity by coordinating and controlling database transactions. The **storage manager**, on the other hand, is tasked with managing the physical storage of data, including data retrieval, indexing, and buffer management.

These components, along with the communication interfaces and the concurrency control mechanisms, work together to create the relational database architecture.

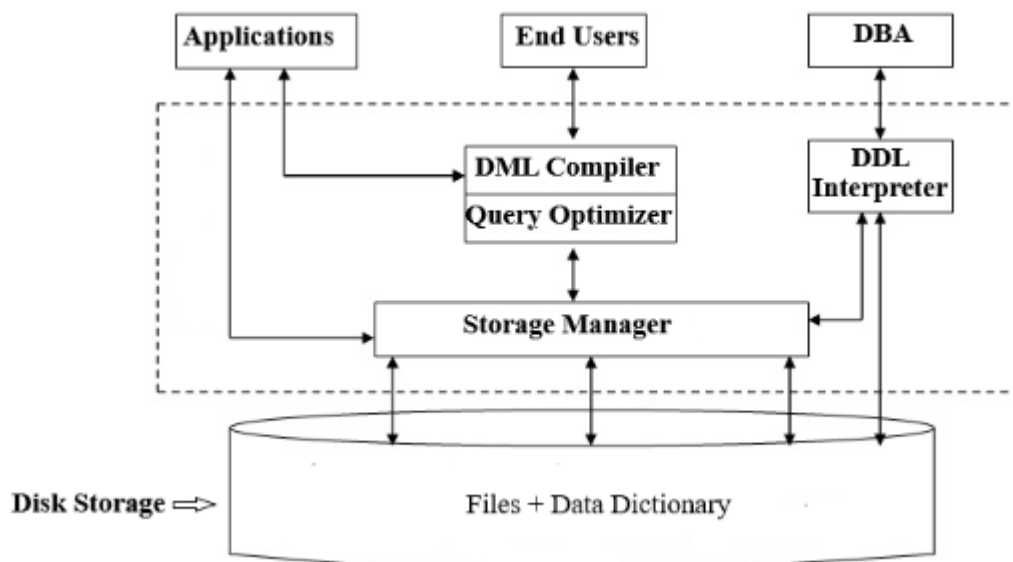
The 5 Key Components of DBMS Architecture:

1. Database Engine:



The database engine is the central component of a DBMS. It is responsible for interpreting and executing user requests, managing data storage and retrieval, and ensuring data integrity and security. It employs various algorithms and data structures to optimize query processing and data access. The database engine also handles concurrent access from multiple users, implementing locking mechanisms to prevent data conflicts. The engine's efficiency directly impacts the overall performance of the database system. It utilizes indexing techniques to speed up data retrieval and employs query optimization strategies to determine the most efficient execution plan for each query. The database engine also manages system resources, such as memory and disk I/O, to maximize throughput and minimize response times.

2. Storage Manager:



The storage manager is responsible for the physical storage of data on the computer's storage devices, such as hard disks or solid-state drives. It handles tasks like file organization, indexing, and buffer management to optimize data access and retrieval.

One of the key functions of the storage manager is to allocate and deallocate disk space as needed. It maintains a free space map, which tracks available storage areas, ensuring efficient use of disk capacity. The storage manager also implements various data access methods, such as sequential, indexed, or hashed, to facilitate quick and efficient data retrieval based on specific query requirements. It also implements mechanisms for data backup and recovery, protecting against data loss due to hardware failures or system crashes.

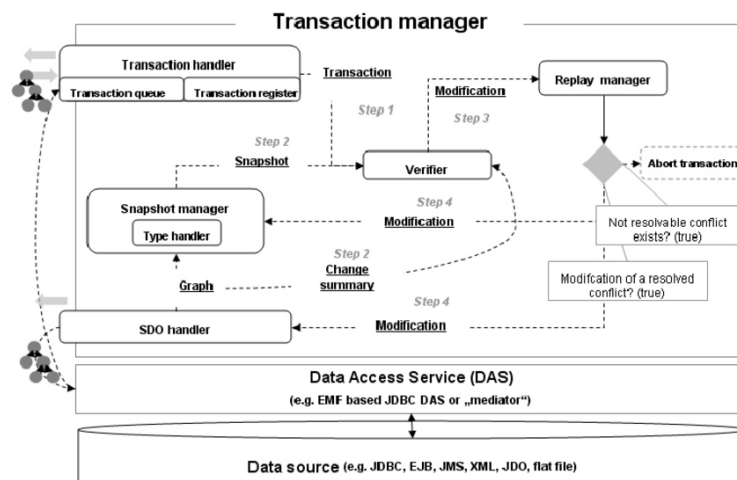
3. Query Processor:

The query processor is responsible for parsing, optimizing, and executing user queries written in a database query language, such as SQL. It analyzes the query, generates an efficient execution plan, and coordinates the various components to retrieve the requested data.

Once the execution plan is created, the steps that **query processor** works in is:

- It interacts with the storage engine to access the necessary data. It employs various algorithms to perform operations like filtering, sorting, and joining data from different tables. The processor also manages query concurrency, ensuring that multiple queries can be executed simultaneously without conflicts.
- During execution, the query processor utilizes system resources such as CPU, memory, and I/O operations efficiently. It may employ techniques like parallel processing for large-scale queries to improve performance. The processor also handles error conditions and exceptions that may occur during query execution, providing meaningful feedback to the user.
- After processing, the query processor formats the results according to the user's specifications and returns them to the client application.

4. Transaction Manager:

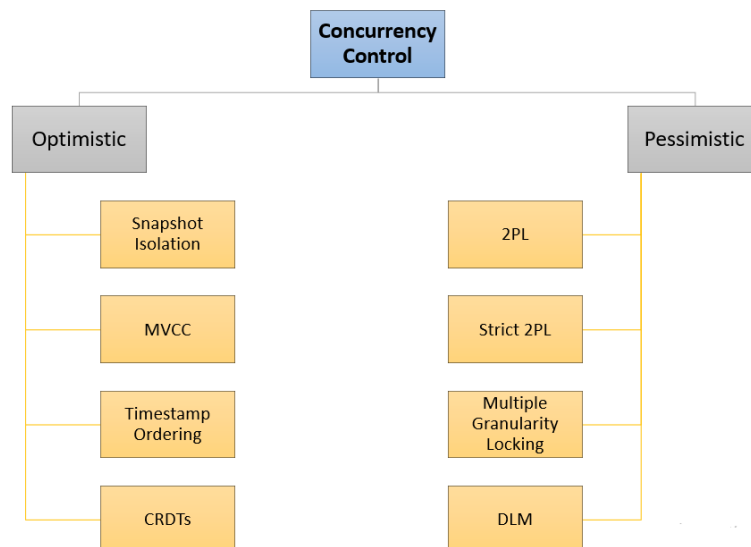


The transaction manager is responsible for ensuring the ACID (Atomicity, Consistency, Isolation, Durability) properties of database transactions. It manages concurrency control, deadlock detection, and recovery mechanisms to maintain data consistency and integrity in distributed systems. The transaction manager coordinates multiple operations across different resources, such as databases or message queues, to ensure they are treated as a single, indivisible unit of work.

One key function of the transaction manager is to implement locking mechanisms, which prevent concurrent access to the same data by multiple transactions. This helps maintain isolation and consistency. It also employs various concurrency control techniques, such as **optimistic** or **pessimistic** locking, to maximize system throughput while preserving data integrity.

In the event of a system failure, the **transaction manager** plays a crucial role in recovery. It maintains transaction logs that record all changes made during a transaction, allowing the system to roll back incomplete transactions or replay committed ones to restore the database to a consistent state.

5. Concurrency Control:



The concurrency control component is responsible for managing simultaneous access to the database by multiple users or processes. It ensures that concurrent transactions do not interfere with each other and that data integrity is maintained, even in the face of concurrent operations.

Key mechanisms employed by concurrency control include

- **Locking:** It prevents multiple transactions from accessing the same data simultaneously
- **Timestamping:** It assigns unique timestamps to transactions to determine their order of execution
- **Multi-version Concurrency Control (MVCC):** MVCC allows multiple versions of data to coexist, enabling read operations to proceed without blocking write operations.

The concurrency control component also implements isolation levels, which define the degree to which one transaction must be isolated from the effects of other concurrent transactions. Common isolation levels include

- Read Uncommitted,
- Read Committed,
- Repeatable Read, and
- Serializable, each offering different trade-offs between consistency and performance.

Additionally, this component handles deadlock detection and resolution, ensuring that transactions do not become indefinitely blocked due to circular dependencies on resources. It may employ techniques such as timeout mechanisms or deadlock detection algorithms to identify and resolve such situations, typically by aborting one or more transactions involved in the deadlock.

Centralized database management system:

Centralized database management system is the system in which all the data is stored and managed in a single unit. This is also known as central computer database system. This system is mostly used in an organization, in any Business Company or in institution to centralize the tasks.

Data can be accessed through a network Local Area Network (LAN) or Wide Area Network (WAN). Mainframe computer is the example of centralized database management system.



Functions of centralized database:

- **Distributed query processing:** The basic function of centralized database management system is to provide facility and give access to all the connected computers which fulfill all requirements requested by any single node.
- **Single central unit :** All the data and information are stored in single centralized database management system. The computer system which fulfills the requirements of all the connected computers is known as server and other computers are known as clients.
- **Transparency:** All the queries are processed in a single computer system also known as server. There is no duplication or irrelevant data stored in this management system. All connected computer has the access to central computer for their query processing and requirement.
- **Scalable:** No of computers can be added in this centralized database management system. These computers are connected to the system through a network.

What are centralized databases used for?

- Centralized databases are often used by organizations to store data that is shared by many users. These are used for storing customer information, inventory data, financial records, and more.
- They can be used by small businesses or large enterprises. When compared to other databases, they offer a number of benefits over other types of databases.
- Centralized database can be accessed by anyone with the proper permissions. This means that multiple people can work on the same data at the same time, which can be a huge time-saver.
- It also allows for better collaboration, as people can easily share data and ideas reliably.
- This is because they are hosted on servers that are designed to be up and running all the time. This means that if one server goes down, the others can still be accessed.

Advantage of centralized database system:

- **Data integrity:** Data is more unified as it stored in single computer system and managed. It is easier to communicate and coordinate to get more reliable and meaningful data.
- **Data redundancy:** Data is centralized and stored in one location only. There is no duplication of data and irrelevancy in data.
- **Data security:** Due to storage of data in centralized computer system, the security of data needs to be stronger. Centralized database management system is more secure and more efficient.

- **Scalability and localization:** New computer systems can be added or removed in centralized database management system more easily.
- **Data portability:** Data can be easily transferred from one computer to another computer because it is stored in centralized database management system.
- **Lesser cost and maintenance:** The centralized database system is cheaper in installation and maintenance than other database management systems and it required single storage system and data can be accessed by all the connected computers.

Disadvantage of centralized database system

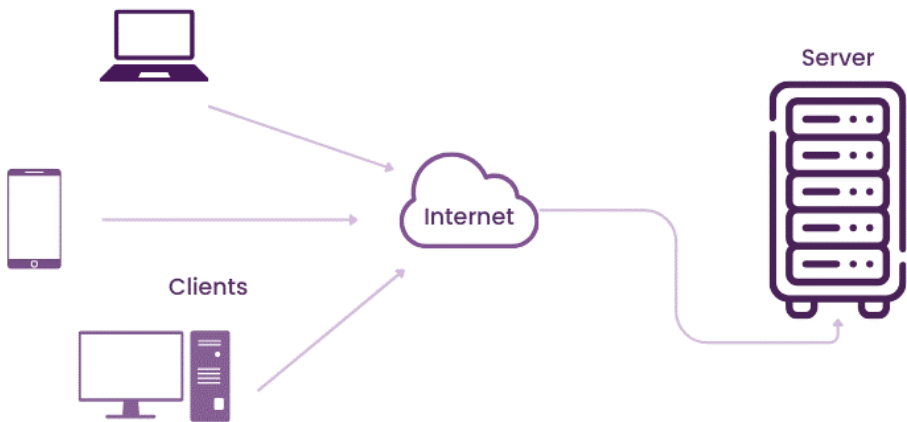
- **Slow processing:** In centralized database management system, data is stored in one location and it access and processing speed is lesser than other management system. It requires more time to access the data from one location.
- **Less efficiency:** If the multiple users try to access and process query toward server simultaneously then it creates problem. The processing speed of the central computer turns into **low**. **These problems may reduce the efficiency.**
- **Loss of data:** In centralized database management system, if any system failure occurs or any data is lost, then it is not recovered.

Difference between centralized and distributed databases:

Aspect	Centralized Databases	Decentralized Databases
Architecture	Client-server model	Peer-to-peer or distributed network
Data Storage	Single central location	Replicated across multiple nodes
Control	Centralized control	Decentralized control
Scalability	Vertical scaling (hardware upgrades)	Horizontal scaling (adding nodes)
Fault Tolerance	Single point of failure	Resilient to node failures
Data Consistency	Robust mechanisms	Challenging to maintain

Aspect	Centralized Databases	Decentralized Databases
Security	Advanced security features	Cryptographic techniques and consensus mechanisms
Management	Simplified administration	Distributed administration
Transparency	Limited transparency	High transparency
Advantages	- Ease of management - Data consistency - Security	- Resilience and fault tolerance - Scalability - Transparency and trust
Disadvantages	- Single point of failure - Scalability challenges - Centralized control	- Complexity - Data consistency challenges - Adoption and integration

What is Client Server Architecture?



Client-Server Architecture is a network model that allows communication and data exchange between different applications over a single or multiple servers.

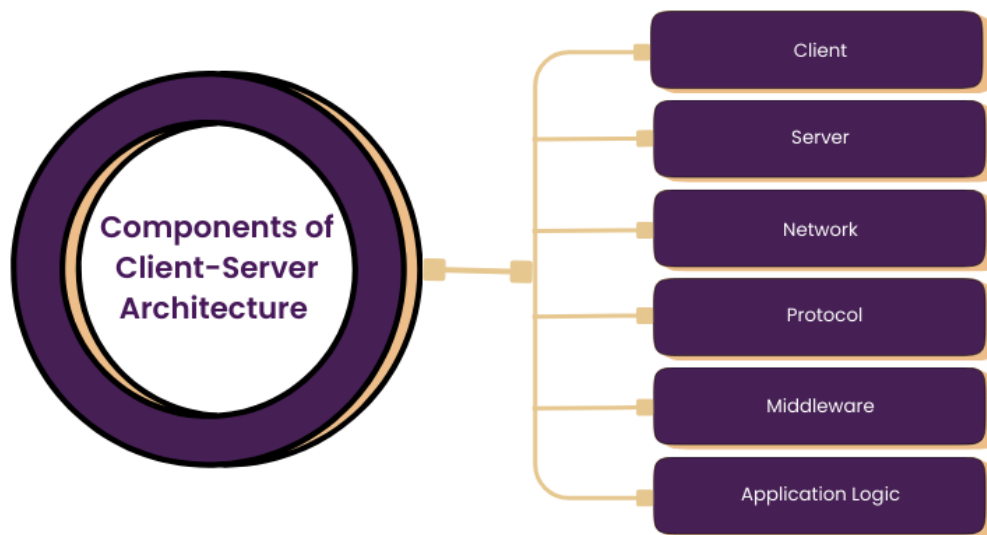
This model divides the system into two parts: the client side and the server side.

a) Client: An application that requests services from the server, such as data retrieval, storage, calculations, or other functions.

b) Server: An application that processes client requests, sends responses, or performs specified actions. The server and client may reside on the same machine or different devices across the network. Effective Communication occurs via predefined protocols like HTTP, FTP, or SMTP.

Client-Server Architecture is widely used in applications such as email, web browsing, online banking, and e-commerce.

Components of Client-Server Architecture



Client-Server Architecture depends on three main components that need to work together for it to function. These components are:

a) Client: A client device or software that requests services from a server. Clients are consumer-facing and often include web browsers, mobile applications, or desktop applications that people can interact with. They communicate with the server to retrieve data, make transactions, or perform other tasks by delegating that responsibility to the server.

b) Server: A server is a computer or program that offers services or solutions to clients over a network. Servers handle processing of client requests, which includes tasks like file storage, database access, and application hosting, along with backend activities like computations, data management, and business logic, significantly reducing what clients need to handle.

c) Network: This serves as the channel through which clients and servers are connected for data transfer between them. Networks range from local area networks (LAN) within a single building to wide area networks (WAN) and the internet, which can span countries. It acts as the intermediary, facilitating the interchange of requests and responses between the clients and servers, which influences the speed and reliability of these interactions.

d) Protocol: Protocols are rules that define how data is exchanged between clients and servers, ensuring communication is orderly, secure, and understandable. Common protocols include HTTP or HTTPS for web services, FTP for file transfers, and SMTP for email. They help bridge communication between different systems, independent of their technology stack.

e) Middleware: Middleware acts as a bridge between client-side and server-side code, enabling them to communicate. It performs tasks such as authentication, load balancing, data translation, and message queuing, simplifying interactions within the Client-Server model by enhancing transaction speed, scalability, and integration.

f) Application Logic: Application logic is the code and processes that determine how a server responds to client requests, involving business rules, Big Data Processing, and workflows on the server side. It ensures the server correctly interprets client requests, performs necessary calculations or data manipulations, and delivers appropriate responses.

Characteristics of Client-Server Architecture

a) The client and server machines can differ greatly in hardware and software requirements and may come from different vendors.

b) The network can scale horizontally by adding more client machines or vertically by upgrading to more powerful servers or a multi-server configuration.

c) A single server can offer multiple services simultaneously, though each service must run its own server program.

d) Both client and server applications interact directly with a transport layer protocol, establishing communication to send and receive information.

e) Both client and server computers require a full stack of protocols, where the transport protocol utilises lower-layer protocols to send and receive individual messages.

How Does Client-Server Architecture Work?

1) In the first step, the client sends a request to the server using the network medium. The request can be a query, a command, or a message.

2) In the second step, the server receives the request and processes it according to its logic and data. The server may access its own resources or other servers to fulfil the request.

3) In the third step, the server sends a response back to the client using the network medium. The response can be Data, an acknowledgement, or an error message.

4) Lastly, the client receives the response and displays it to the user or performs further actions based on it.

Applications:

1) Email Servers: Email has evolved into the primary communication method for businesses due to its speed and convenience. Various server components work together to deliver emails between users across different mail servers.

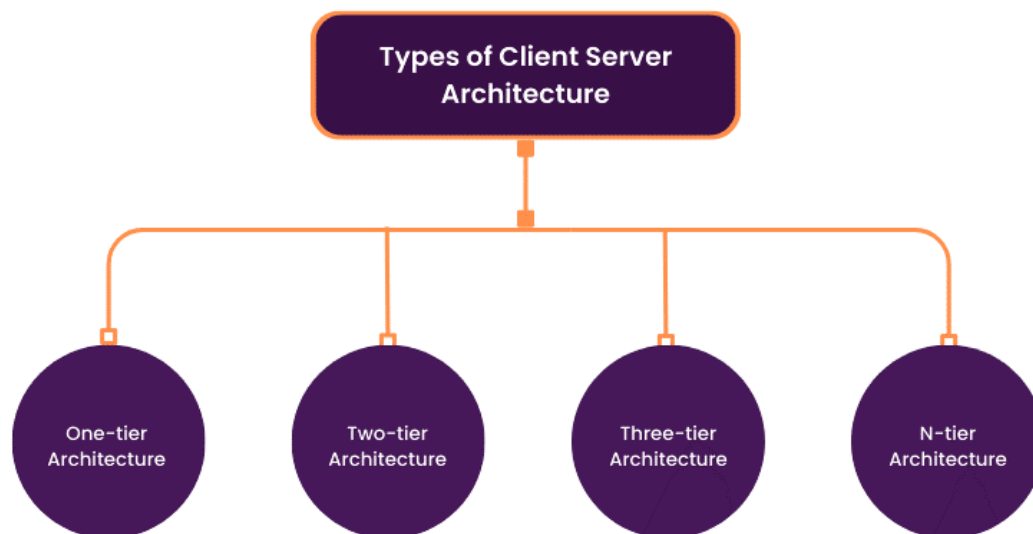
2) File Servers: When saving documents on services like Google Docs or Microsoft Office, you're interacting with file servers. These servers store data centrally and allow multiple clients to access it.

3) Web Servers: These high-powered servers host websites, which web clients access through DNS or an IP address. Here's a simplified process:

- a) A user enters a URL in the browser.
- b) The browser requests the IP address from the Domain Name System (DNS) .
- c) The DNS server provides the IP address to the browser.
- d) The browser sends an HTTPS or HTTP request to the web server.
- e) The server sends back the requested files.
- f) The user retrieves the files, and the process continues as needed.

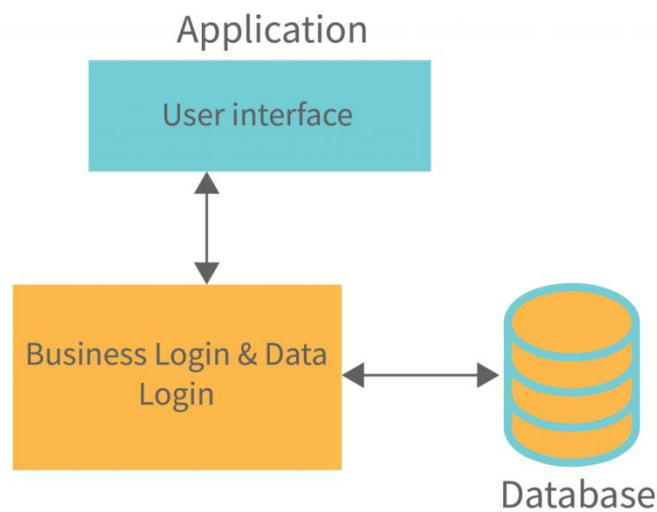
These examples highlight how Client-Server Architecture forms the foundation of many digital services we rely on daily.

Types of Client-Server Architecture



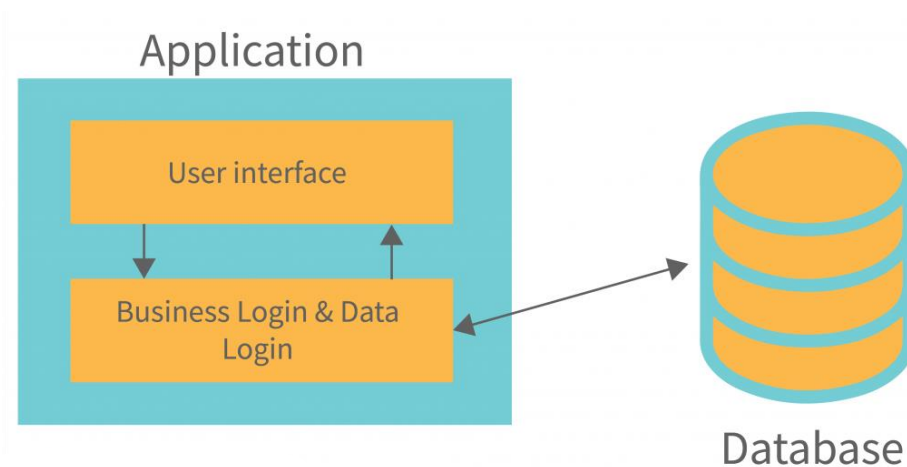
There are different types of Client-Server Architecture, depending on how many tiers or layers are involved in the communication process. Some of the common types are:

1) One-tier Architecture



A self-contained application on a single platform. In one-tier architecture, the client, server, and database are all on the same machine. The client handles user interaction and business logic, the server provides services like data storage and processing, and the database manages data. While simple and popular for small apps, this architecture is rarely used in production because it doesn't meet most system requirements.

2) Two-tier Architecture

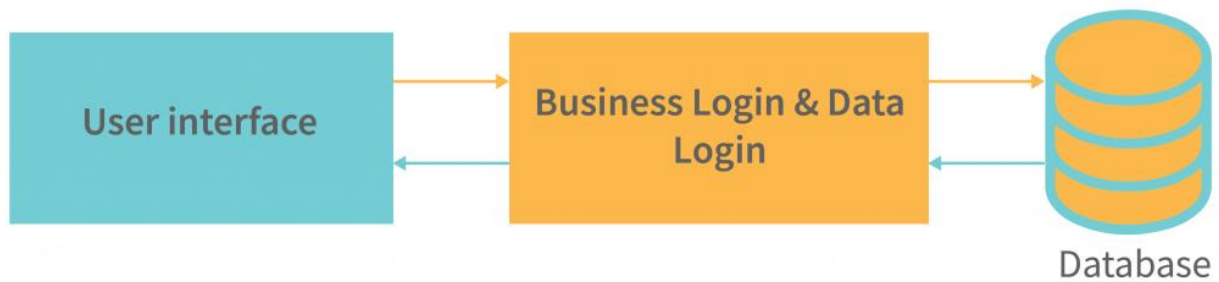


This basic Client-Server Architecture involves direct communication between the client and server without an intermediate layer. The client manages the user interface (UI) and business logic, while the server handles data storage and processing. An example is a web browser requesting pages from a web server, which responds with HTML files. It's easy to implement but has drawbacks like low scalability, high network traffic, and security risks.

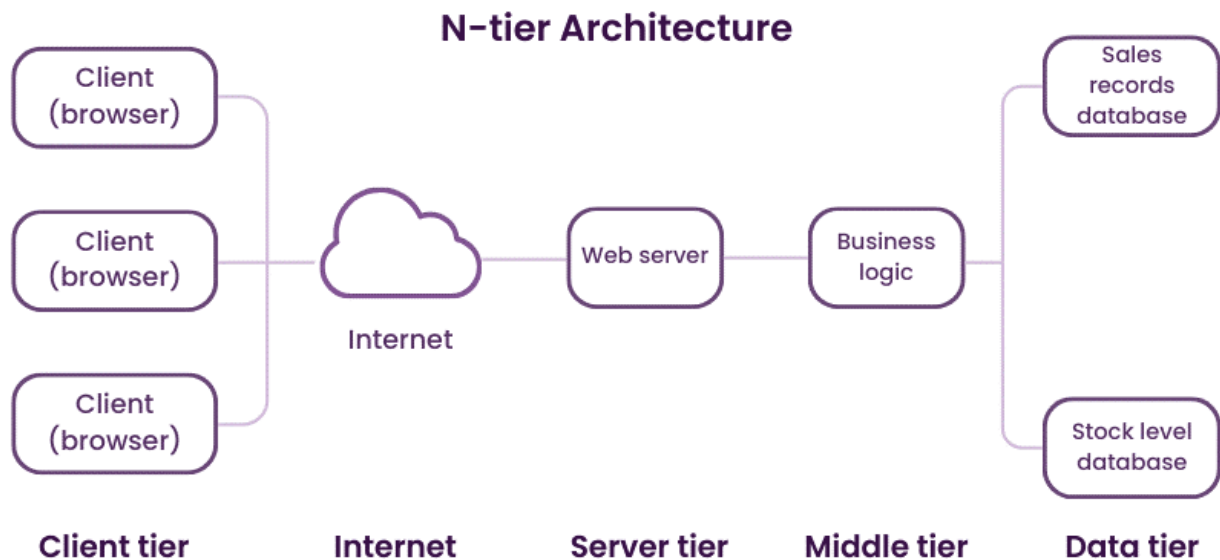
Three-tier Architecture

A more complex client-server setup with an intermediate layer (usually middleware or an application server) that handles business logic, acting as a bridge between the client and server. The client deals with the user interface (UI), while the server manages data storage. An example is an online banking system where the

client is a web browser, the middleware checks transactions, and the server stores account data. This architecture improves scalability, performance, and security but increases complexity and cost.



N-tier Architecture



A more flexible Client-Server Architecture with more than three tiers, allowing greater scalability, flexibility, and modularity. Each tier can be distributed across different machines or networks and updated independently. An example is an e-commerce system with a web browser displaying the product catalog, a web server handling HTTP requests, an application server processing business logic, and a database storing product information. While suitable for complex systems, it requires more resources and management.

In addition to these types, there are also some variations or subtypes of Client-Server Architecture, such as:

Peer-to-peer Architecture

This Client-Server Architecture allows each node to act as both a client and a server, with no central authority or hierarchy. Nodes can directly request or provide services to each other. An example is BitTorrent, a file-sharing system where nodes can download or upload files from each other without a

central server, enhancing reliability, availability, and resilience. However, it presents challenges like security, coordination, and quality control.

Thin-client Architecture

In this architecture, the client has minimal functionality, relying on the server for processing and data storage. The client handles only the user interface (UI) and sends inputs to the server, which performs all computations and returns the results. An example is Google Docs, where the client is a web browser accessing the document editor while the server stores and updates data in the cloud. This architecture reduces client-side hardware costs, maintenance, and security risks but increases network dependency, bandwidth usage, and server load.

Fat-client Architecture

In a fat-client setup, the client is highly functional, performing most processing and data storage locally, and only occasionally communicates with the server for synchronisation or backup. An example is an offline mobile app like Evernote, where the client runs the app and stores notes locally, syncing with the cloud when connected. This architecture boosts performance, responsiveness, and availability on the client side but increases hardware requirements, software complexity, and risks of data inconsistency.

Advantages

- 1) It is centralised, which means that all data and services are stored and managed in a single place. This makes it easier to maintain, update, and secure the system.
- 2) It is cost-efficient, as it requires less hardware and software resources for the client side. The client only needs a network connection and an application or web browser to access the server.
- 3) It has high performance and low latency, as the server can handle many requests from many clients simultaneously and efficiently.

Disadvantages

- 1) It has limited scalability, as it depends on the capacity and availability of the server. If the server is overloaded or fails, the system may not function properly or at all.
- 2) It has high network dependency, as it relies on the network connection between the client and the server. If the network is slow or disrupted, the system may experience delays or errors.
- 3) It has complex architecture, as it involves multiple components and layers that need to be designed, implemented, and coordinated. The system may also face challenges such as security, synchronisation, and compatibility.

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Parallel database management system (DBMS)

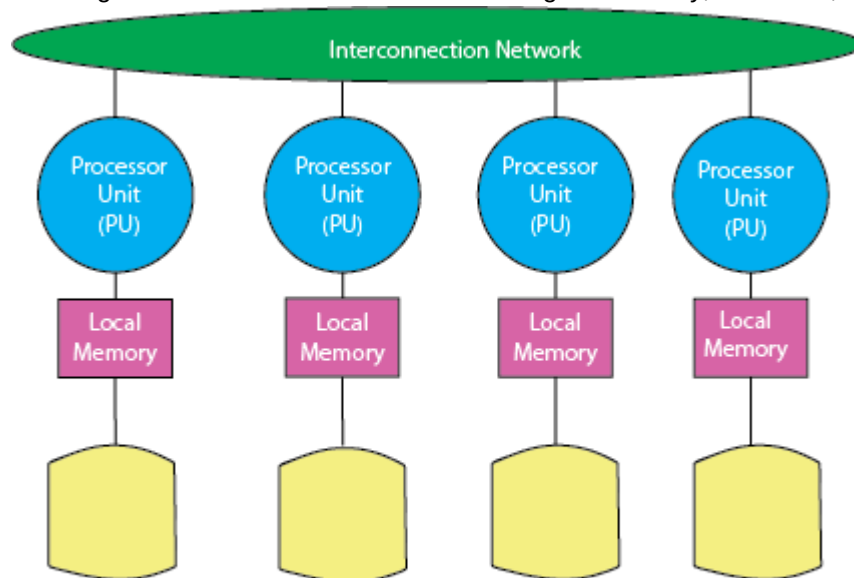
A parallel database management system (DBMS) is a database management system that works on several processors or CPUs and is primarily designed to execute query operations in parallel whenever possible. The parallel DBMS connects several smaller machines to obtain the same throughput as a single large system. Parallelism is utilized to support query speedup, in which extra resources, like processors and drives, are made available. Scale-up is another application of parallelism, in which increasing workloads are managed without increasing response time by raising the degree of parallelism.

There are three types of architectures for the parallel database management system (DBMS), Here I am going to describe them,

Shared Nothing Architecture:

Shared nothing architecture is a distributed database management system. It is also called the shared-nothing architecture because the nodes are not connected to each other.

The nodes are generally independent, horizontally partitioned, and synchronized by messages that are passed between them. Nodes can be added or removed from the system at any time without affecting the other nodes in the system. This type of architecture can be used to create large-scale systems that are capable of handling an enormous amount of data with high availability, resilience, and scalability.



Shared-Nothing Architecture

Advantages of Shared nothing disk system

- Number of processors and disk can be connected as per the requirement in share nothing disk system.
- Shared nothing disk system can support for many processor, which makes the system more scalable.

Disadvantages of Shared nothing disk system

- Data partitioning is required in shared nothing disk system.

- Cost of communication for accessing local disk is much higher.

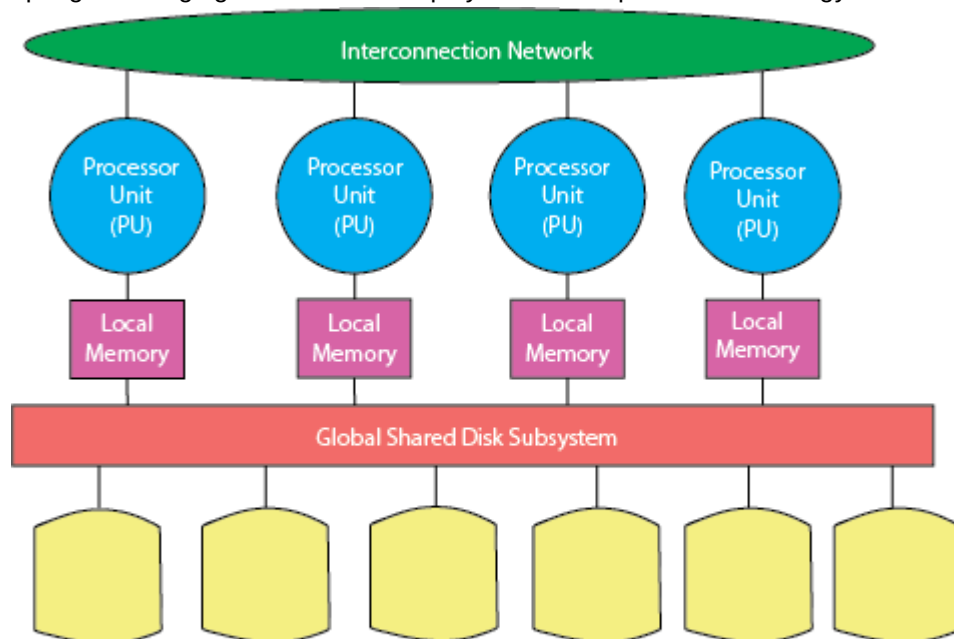
Applications of Shared nothing disk system

- Tera data database machine.
- The Grace and Gamma research prototypes.

Shared Disk Architecture:

The concept of shared ownership of the complete database between RDBMS servers, each of which is executing on a node of a distributed memory system, is implemented using shared-disk architecture. Each RDBMS server can read, write, update, and remove data from the same shared database, necessitating the need for a distributed lock manager in the system (DLM)

The discs are accessible from all cluster nodes in this architecture. This architecture is capable of quickly adapting to changing workloads. It employs a robust optimization strategy.



Distributed memory shared-disk architecture

Advantages of Shared Disk System

- Fault tolerance is achieved using shared disk system.
Fault tolerance: If a processor or its memory fails, the other processor can complete the task. This is called as fault tolerance.

Disadvantage of Shared Disk System

- Shared disk system has limited scalability as large amount of data travels through the interconnection channel.
- If more processors are added the existing processors are slowed down.

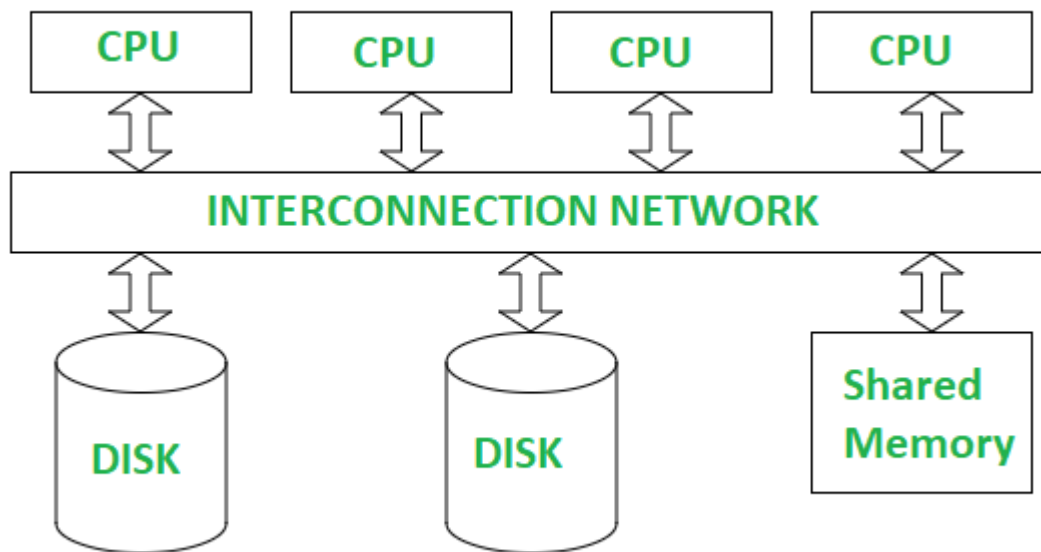
Applications of Shared Disk System

Digital Equipment Corporation(DEC): **DEC cluster running relational databases use the shared disk system and now owned by Oracle.**

Shared-Memory Architecture:

Shared-memory or shared-everything style is the standard technique of building an RDBMS on SMP hardware. It is quite straightforward to construct and has proven highly effective up to the point when it runs into the shared-everything architecture's scalability limitations. The major benefit of this approach is that a single RDBMS server can most likely use all CPUs, access all memory, and access the whole database, giving the client a consistent single system image.

The DBMS assumes that many database components executing SQL queries interact with each other via shared memory in shared-memory SMP systems. All CPUs have access to all data on the local disc, which is partitioned.



Advantages of Shared memory system

- Data is easily accessible to any processor.
- One processor can send message to other efficiently.

Disadvantages of Shared memory system

- Waiting time of processors is increased due to more number of processors.
- Bandwidth problem.

DATA WAREHOUSE:

Data is an individual unit that contains raw materials which do not carry any specific meaning.

Information is a group of data that collectively carries a logical meaning. Data doesn't depend on information.

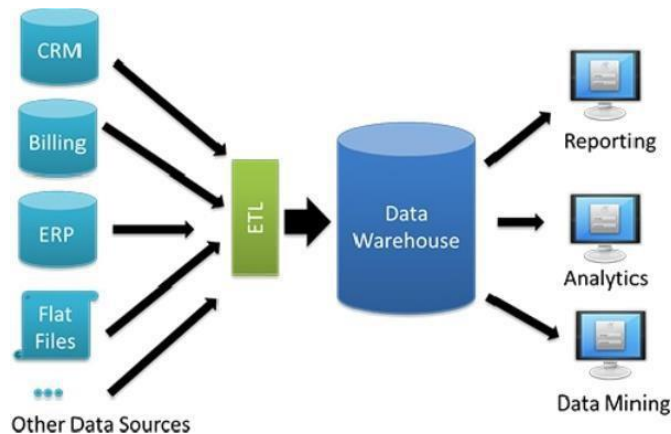
Information depends on data. Data is

measured in bits and bytes.

Information is measured in meaningful units like time, quantity, etc.

Data Warehouse:

Data warehouse is like a relational database designed for analytical needs. It functions on the basis of OLAP (Online Analytical Processing). It is a central location where consolidated data from multiple locations (databases) are stored.



Data warehousing:

Data warehousing is the act of organizing & storing data in a way so as to make its retrieval efficient and insightful. It is also called as the process of transforming data into information.

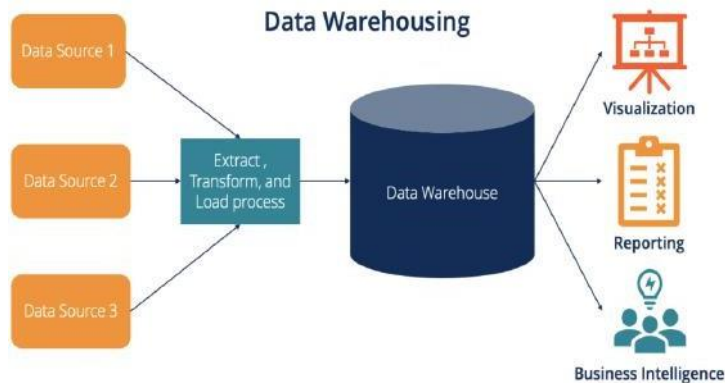


Fig: Data warehousing Process

Data Warehouse Characteristics:

A Data warehouse is a subject-oriented, integrated, time variant and non-volatile collection of data in support of management's decision making process.

Subject-oriented:

A Data warehouse can be used to analyze a particular subject area

Ex: "Sales" can be particular subject

Integrated:

A Data warehouse integrates data from multiple data sources.

Time Variant:

Historical data is kept in a data warehouse.

Ex: one can retrieve data from 3 months, 6months, 12 months or even older data from a data warehouse. This contrasts with a transactions system, where often only the most recent data is kept.

Non-Volatile:

Once data is in the data warehouse, it will not change. So historical data in a data warehouse should never be altered.

Data warehouse Architecture:

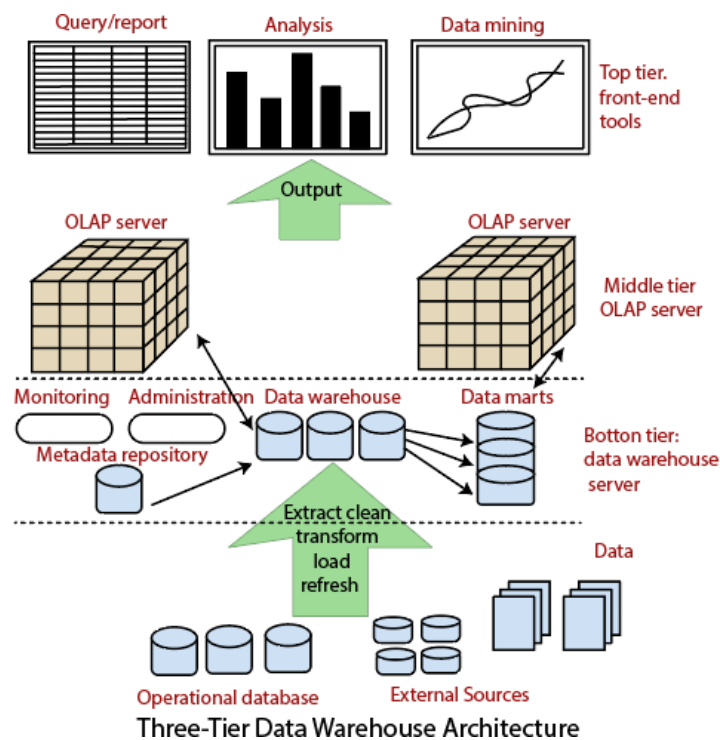


Fig: Data ware housing Architecture

Data warehouses often adopt a three-tier architecture

- The bottom tier is a **warehouse database server** that is almost always a relational database system. Back-end tools and utilities are used to feed the data into the bottom tier from operational database or other external sources. These tools and utilities perform data extraction, cleaning and transformation(ex. To merge similar data from different sources into a unified format), as well as load and refresh functions to update the data warehouse. The data are extracted using application program interfaces known as gateways.A gateway is supported by the underlying DBMS and allows client programs to generate SQL code to be executed at a server.

Examples of gateways include **ODBC**(Open Database Connection) and **OLEDB**(Open Linking

and Embedding for Databases) by Microsoft and **JDBC**(Java Database Connection).This tier also contains a metadata repository, which stores information about the data warehouse and its contents.

The middle tier is an **OLAP** server that is typically implemented using either

(a) a relational OLAP(**ROLAP**) model, that is an extended relational DBMS that maps operations on multidimensional data to standard relational operations, or a multidimensional OLAP(**MOLAP**) model that is a special-purpose server that directly implements multidimensional data and operations.

The top tier is a front end client layer, which contains query and reporting tools, analysis tools and data mining tools(ex: trend analysis, prediction....)

Multi-dimensional Data Model:

- A multidimensional model views data in the form of a data-cube.
- When data is grouped or combined in multidimensional matrices called Data Cubes.
- A data cube enables data to be modeled and viewed in multiple dimensions. It is defined by dimensions and facts.
- A multidimensional data model is organized around a central theme, for example, sales. This theme is represented by a fact table. Facts are numerical measures.
- The fact table contains the names of the facts or measures of the related dimensional tables.

FACT VS DIMENSION

Fact/Measure(What you want to analyse is your fact)

Ex: What is My sales, What is my profit, What is my customers preferences. Dimensions(By Which you want to Analyze is your Dimensions)
Sales By Location/Product/Period

Total Profit By Location/Product/Period

- These Dimensions allow the store to keep track of things like monthly sales of items and branches and locations at which the items were sold.
- Each dimension may have a table associated with it called a **dimension table**, which further describes the dimension.

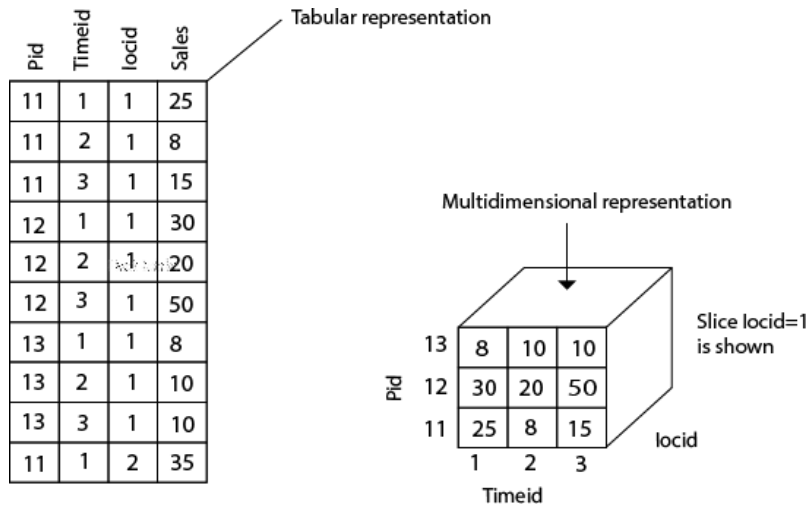


Fig: Multidimensional Representation

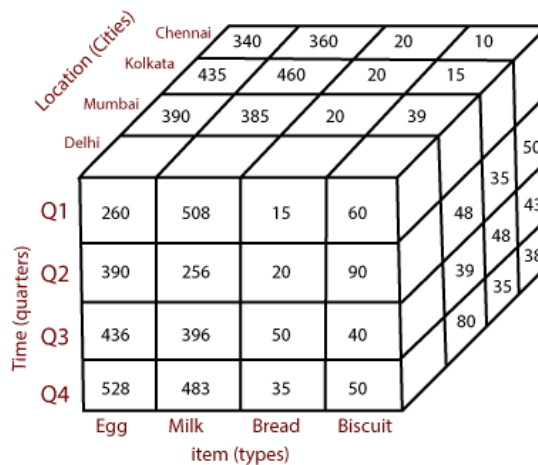
- Consider the data of a shop for items sold per quarter in the city of Delhi. The data is shown in the table.
- In this 2D representation, the sales for Delhi are shown for the time dimension (organized in quarters) and the item dimension (classified according to the types of an item sold).
- The fact or measure displayed in rupee_sold (in thousands).

Location="Delhi"				
Time (quarter)	item (type)			
	Egg	Milk	Bread	Biscuit
Q1	260	508	15	60
Q2	390	256	20	90
Q3	436	396	50	40
Q4	528	483	35	50

Now, if we want to view the sales data with a third dimension, For example, suppose the data according to time and item, as well as the location is considered for the cities Chennai, Kolkata, Mumbai, and Delhi. These 3D data are shown in the table. The 3D data of the table are represented as a series of 2D tables.

	Location="Chennai"				Location="Kolkata"				Location="Mumbai"				Location="Delhi"			
	item				item				item				item			
Time	Egg	Milk	Bread	Biscuit	Egg	Milk	Bread	Biscuit	Egg	Milk	Bread	Biscuit	Egg	Milk	Bread	Biscuit
Q1	340	360	20	10	435	460	20	15	390	385	20	39	260	508	15	60
Q2	490	490	16	50	389	385	45	35	463	366	25	48	390	256	20	90
Q3	680	583	46	43	684	490	39	48	568	594	36	39	436	396	50	40
Q4	535	694	39	38	335	365	83	35	338	484	48	80	528	483	35	50

Conceptually, it may also be represented by the same data in the form of a 3D data cube, as shown in fig:



What is Schema?

- Schema is a logical description of the entire database.
- It includes the name and description of records of all record types including all associated data-items and aggregates.
- Much like a database, a data warehouse also requires to maintain a schema.
- A database uses relational model, while a data warehouse uses Star, Snowflake, and Fact Constellation schema.
- Modeling data warehouses: dimensions & measures

- Star schema: A fact table in the middle connected to a set of dimension tables
- Snowflake schema: A refinement of star schema where some dimensional hierarchy is normalized into a set of smaller dimension tables, forming a shape similar to snowflake
- Fact constellations: Multiple fact tables share dimension tables, viewed as a collection of stars, therefore called galaxy schema or fact constellation

Star Schema:

- A star schema is the elementary form of a dimensional model, in which data are organized into **facts** and **dimensions**.
- A fact is an event that is counted or measured, such as a sale or log in. A dimension includes reference data about the fact, such as date, item, or customer.
- A star schema is a relational schema where a relational schema whose design represents a multidimensional data model.
- The star schema is the explicit data warehouse schema. It is known as **star schema** because the entity-relationship diagram of this schemas simulates a star, with points, diverge from a central table.
- The center of the schema consists of a large fact table, and the points of the star are the dimension tables.

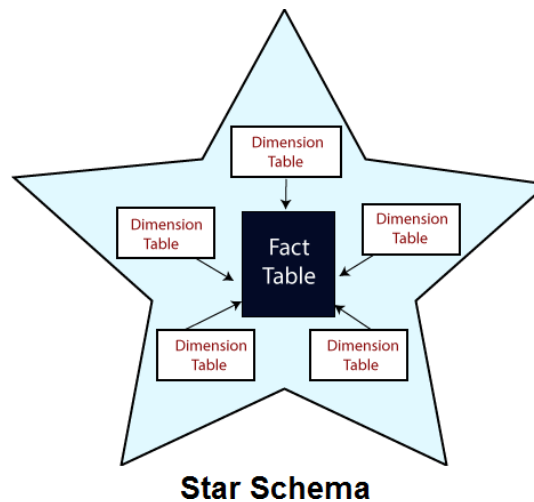
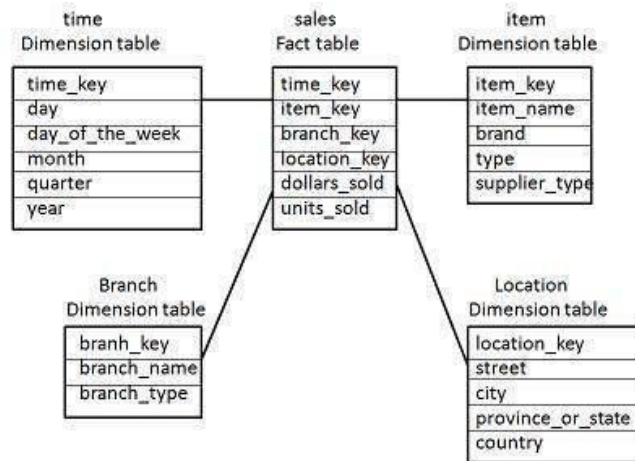


Fig: Star Schema Representation

Star Schema:

- Each dimension in a star schema is represented with only one-dimension table.
- This dimension table contains the set of attributes.
- The following diagram shows the sales data of a company with respect to the four dimensions, namely time, item, branch, and location.



- There is a fact table at the center. It contains the keys to each of four dimensions.
- The fact table also contains the attributes, namely dollars sold and units sold.
- Each dimension has only one dimension table and each table holds a set of attributes. For example, the location dimension table contains the attribute set {location_key, street, city, province_or_state, country}. This constraint may cause data redundancy. For example, "Vancouver" and "Victoria" both the cities are in the Canadian province of British Columbia. The entries for such cities may cause data redundancy along the attributes province_or_state and country.

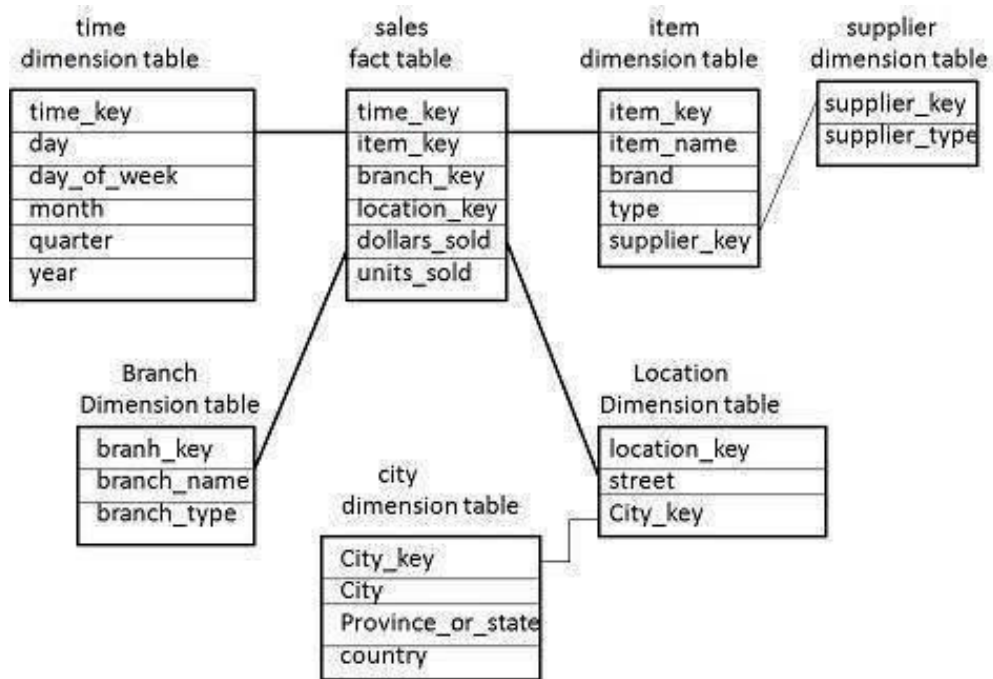
Characteristics of Star Schema:

- Every dimension in a star schema is represented with the only one-dimension table.
- The dimension table should contain the set of attributes.
- The dimension table is joined to the fact table using a foreign key
- The dimension table are not joined to each other
- Fact table would contain key and measure
- The Star schema is easy to understand and provides optimal disk usage.

- The dimension tables are not normalized. For instance, in the above figure, Country_ID does not have Country lookup table as an OLTP design would have.
- The schema is widely supported by BI Tools.
- Advantages:
 - (i) Simplest and Easiest
 - (ii) It optimizes navigation through database
 - (iii) Most suitable for Query Processing

Snowflake Schema:

- Some dimension tables in the Snowflake schema are normalized.
- The normalization splits up the data into additional tables.
- Unlike Star schema, the dimensions table in a snowflake schema are normalized.
- For example, the item dimension table in star schema is normalized and split into two dimension tables, namely item and supplier table.



- Now the item dimension table contains the attributes item_key, item_name, type, brand, and supplier-key.
- The supplier key is linked to the supplier dimension table. The supplier dimension table contains the attributes supplier_key and supplier_type.

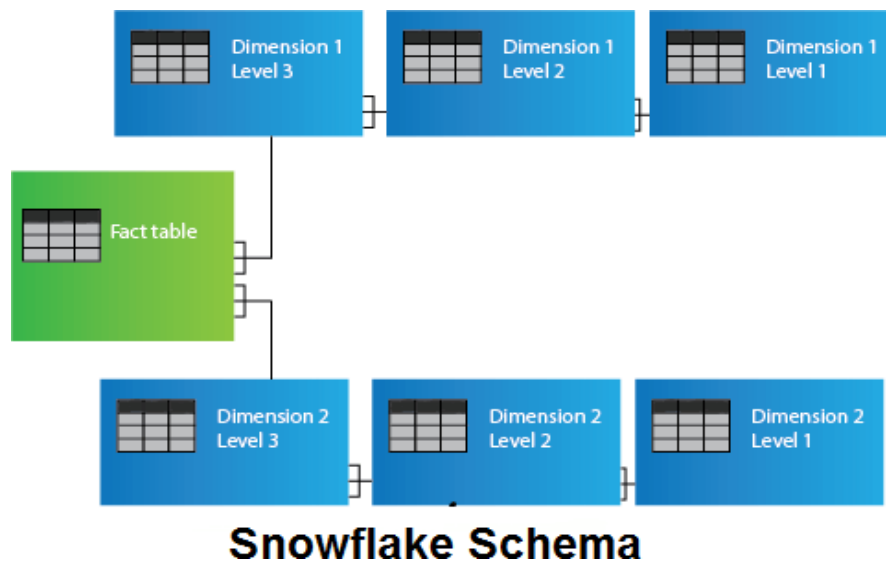
Note : Due to normalization in the Snowflake schema, the redundancy is reduced and therefore, it becomes easy to maintain and the save storage space.



Fig: Snowflake image

A snowflake schemas can have any number of dimension, and each dimension can have any number of levels.

The following diagram shows a snowflake schema with two dimensions, each having three levels.



Advantages:

(i) Less redundancies due to normalization Dimension Tables.

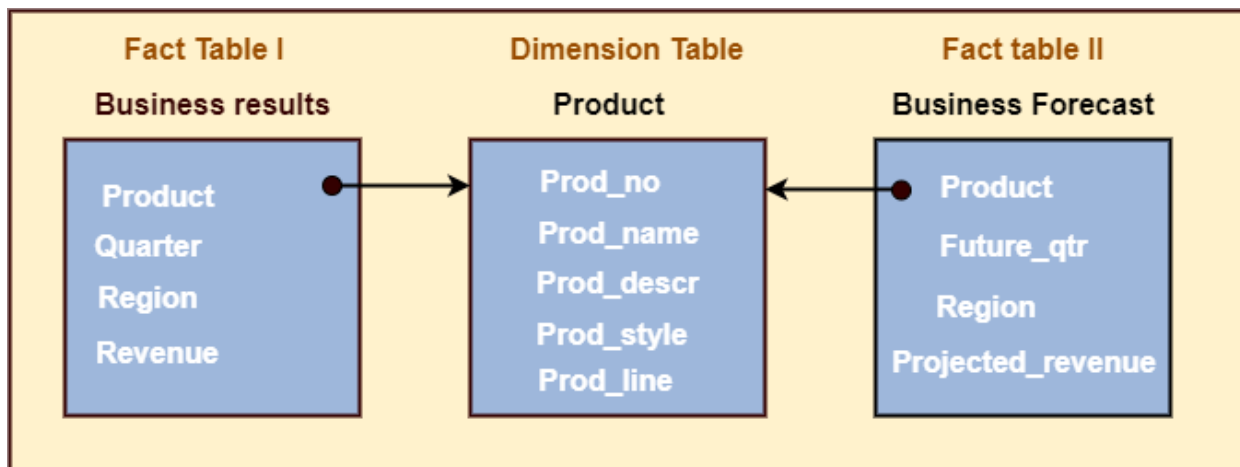
(ii) Dimension Tables are easier to update.

Disadvantages:

It is complex schema when compared to star schema.

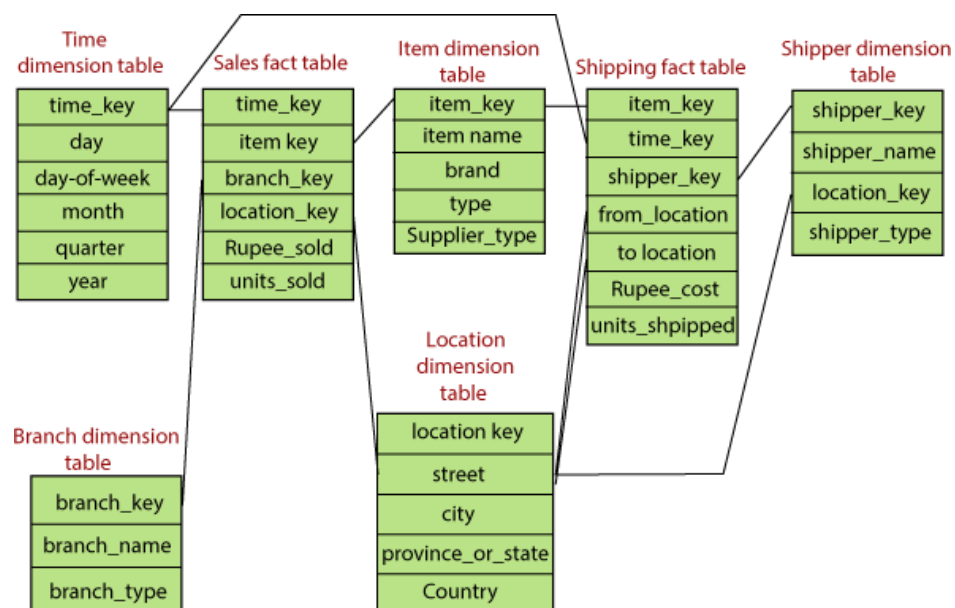
Fact Constellation Schema:

- A Fact constellation means two or more fact tables sharing one or more dimensions. It is also called **Galaxy schema**.
- Fact Constellation Schema describes a logical structure of data warehouse or data mart. Fact Constellation Schema can design with a collection of de-normalized FACT, Shared, and Conformed Dimension tables.



FACT Constellation Schema

A fact constellation schema is shown in the figure below.



- This schema defines two fact tables, sales, and shipping. Sales are treated along four dimensions, namely, time, item, branch, and location.
- The schema contains a fact table for sales that includes keys to each of the four dimensions, along with two measures: Rupee_sold and units_sold.
- The shipping table has five dimensions, or keys: item_key, time_key, shipper_key, from_location, and to_location, and two measures: Rupee_cost and units_shipped.
- It is also possible to share dimension tables between fact tables. For example, time, item, and location dimension tables are shared between the sales and shipping fact table.

Disadvantages:

- (i) Complex due to multiple fact tables
- (ii) It is difficult to manage
- (iii) Dimension Tables are very large.

OLAP OPERATIONS:

- In the multidimensional model, the records are organized into various dimensions, and each dimension includes multiple levels of abstraction described by concept hierarchies.
- This organization support users with the flexibility to view data from various perspectives.
- A number of OLAP data cube operation exist to demonstrate these different views, allowing interactive queries and search of the record at hand. Hence, OLAP supports a user-friendly environment for interactive data analysis.
- Consider the OLAP operations which are to be performed on multidimensional data.
- The data cubes for sales of a shop. The cube contains the dimensions, location, and time and item, where the **location** is aggregated with regard to city values, **time** is aggregated with respect to quarters, and an **item** is aggregated with respect to item types.

OLAP having 5 different operations

- (i) Roll-up
- (ii) Drill-down
- (iii) Slice
- (iv) Dice
- (v) Pivot

Roll-up:

- The roll-up operation performs aggregation on a data cube, by climbing down concept hierarchies, i.e., dimension reduction. Roll-up is like **zooming-out** on the data cubes.
- It is also known as drill-up or aggregation operation
- Figure shows the result of roll-up operations performed on the dimension location. The hierarchy for the location is defined as the Order Street, city, province, or state, country.
- The roll-up operation aggregates the data by ascending the location hierarchy from the level of the city to the level of the country.
- When a roll-up is performed by dimensions reduction, one or more dimensions are removed from the cube.
- For example, consider a sales data cube having two dimensions, location and time. Roll- up may be performed by removing, the time dimensions, appearing in an aggregation of the total sales by location, relatively than by location and by time.

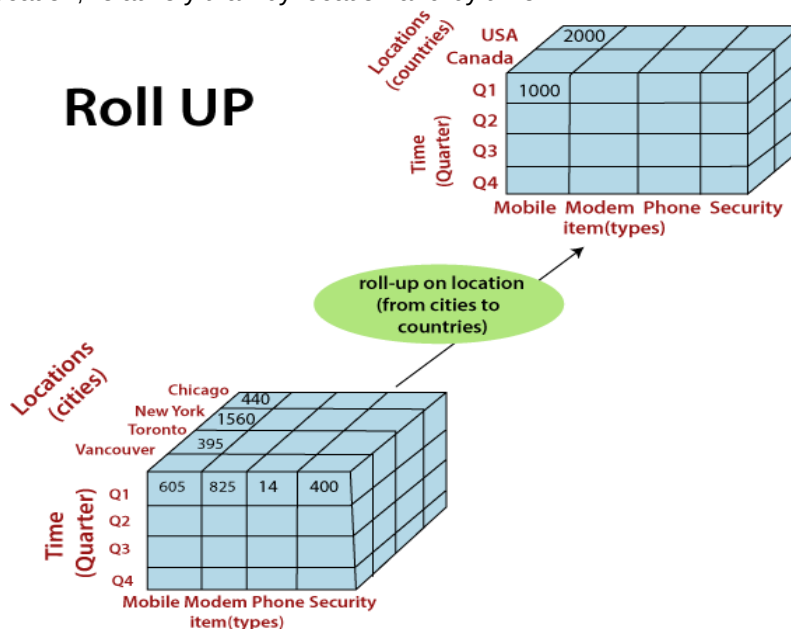


Fig: Roll-up operation on Data Cube

Drill-Down

- The drill-down operation is the reverse operation of **roll-up**.
- It is also called roll-down operation.
- Drill-down is like **zooming-in** on the data cube.
- It navigates from less detailed record to more detailed data. Drill-down can be performed by either **stepping down** a concept hierarchy for a dimension or adding additional dimensions.
- Figure shows a drill-down operation performed on the dimension time by stepping down a

concept hierarchy which is defined as day, month, quarter, and year.

- Drill-down appears by descending the time hierarchy from the level of the quarter to a more detailed level of the month.
- Because a drill-down adds more details to the given data, it can also be performed by adding a new dimension to a cube.

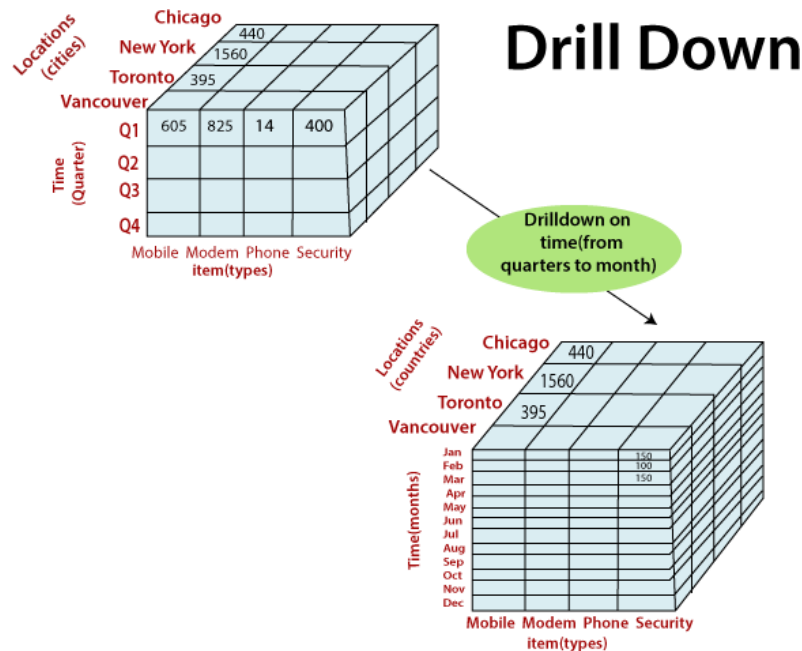


Fig: Drill-down operation

Slice:

- A **slice** is a subset of the cubes corresponding to a single value for one or more members of the dimension.
- The slice operation provides a new sub cube from one particular dimension in a given cube.
- For example, a slice operation is executed when the customer wants a selection on one dimension of a three-dimensional cube resulting in a two-dimensional site. So, the Slice operations perform a selection on one dimension of the given cube, thus resulting in a sub cube.
- Here Slice is functioning for the dimensions "time" using the criterion time = "Q1".
- It will form a new sub-cubes by selecting one or more dimensions.

Slice

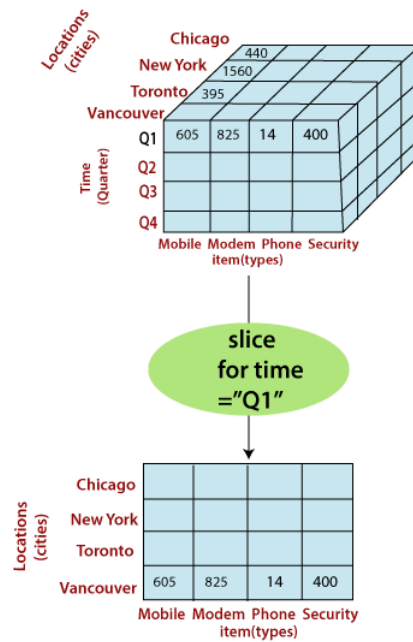


Fig: Slice operation

Dice:

- The dice operation describes a sub cube by operating a selection on two or more dimension.

Dice

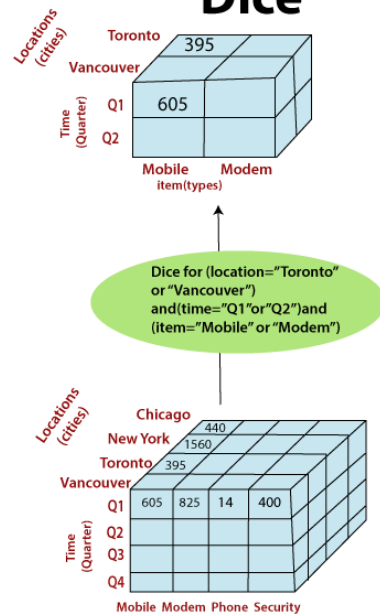


Fig: Dice operation

- The dice operation on the cubes based on the following selection criteria involves three dimensions.

(location = "Toronto" or "Vancouver") (time =
"Q1" or "Q2")
(item =" Mobile" or "Modem")

Pivot:

- The pivot operation is also called a rotation.
- Pivot is a visualization operations which rotates the data axes in view to provide an alternative presentation of the data.
- It may contain swapping the rows and columns or moving one of the row-dimensions into the column dimensions.

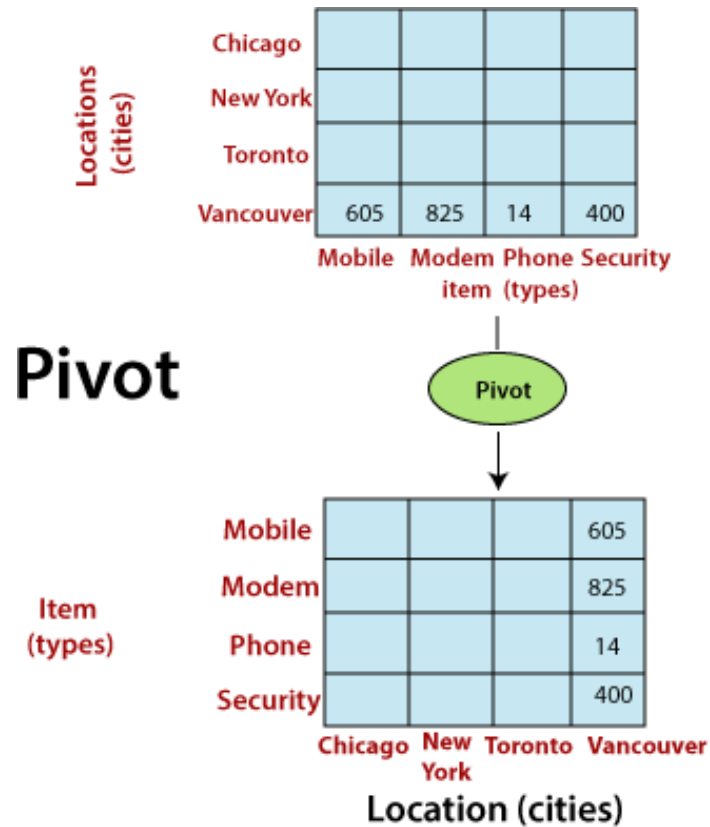


Fig: Pivot Operation

Difference Between OLAP and OLTP:

CATEGORY	OLTP	OLAP
QUERY TYPES	It process simple queries.	It process complex queries.
CHARACTERISTICS	Manages large amount of small transactions.	Manages large volume of data.
QUERY	Processes Insert, Update and Delete queries.	Processes select queries.
RESPONSE TIME	The Responses time is in miliseconds.	Response time is in seconds or minutes.
DATABASE	Uses traditional database.	Uses Data warehouse.
SOURCE	OLTP is the source of data	Various OLTP databases becomes source of data for OLAP.

Types of Data Warehouse:

There are three main types of DWH. Each has its specific role in data management operations.

1. Enterprise Data Warehouse

Enterprise data warehouse (EDW) serves as a central or main database to facilitate decision-making throughout the enterprise. Key benefits of having an EDW include access to cross-organizational information, the ability to run complex queries, and the enablement of enriched, far-sighted insights for data-driven decisions and early risk assessment.

2. ODS (Operational Data Store)

In ODS, the DWH refreshes in real-time. Therefore, organizations often used it for routine enterprise activities, such as storing records of the employees. Business processes also use ODS as a source for providing data to the EDW.

3. Data Mart

It is a subset of a DWH that supports a particular department, region, or business unit. Consider this: You have multiple departments, including sales, marketing, product development, etc. Each department will have a central repository where it stores data. This repository is called a data mart. The EDW stores the data from the data mart in the ODS on a daily/weekly (or as configured) basis. The ODS acts as a staging area for data integration. It then sends the data to the EDW to store it and use it for BI purposes.

DATAWAREHOUSE COMPONENTS:

The data warehouse is based on an RDBMS server which is a central information repository that is surrounded by some key components to make the entire environment functional, manageable and accessible

There are mainly five components of Data Warehouse:

DATA WAREHOUSE DATABASE:

The central database is the foundation of the data warehousing environment. This database is implemented on the RDBMS technology. Although, this kind of implementation is constrained by the fact that traditional RDBMS system is optimized for transactional database processing and not for data warehousing. For instance, ad-hoc query, multi-table joins, aggregates are resource intensive and slow down performance.

Hence, alternative approaches to Database are used as listed below-

In a data warehouse, relational databases are deployed in parallel to allow for scalability. Parallel relational databases also allow shared memory or shared nothing model on various multiprocessor configurations or massively parallel processors.

New index structures are used to bypass relational table scan and improve speed.

Use of multidimensional database (MDDBs) to overcome any limitations which are placed because of the relational data model. Example: Essbase from Oracle.

SOURCING, ACQUISITION, CLEAN-UP AND TRANSFORMATION TOOLS (ETL):

The data sourcing, transformation, and migration tools are used for performing all the conversions, summarizations, and all the changes needed to transform data into a unified format in the data warehouse. They are also called Extract, Transform and Load (ETL) Tools.

These Extract, Transform, and Load tools may generate cron jobs, background jobs, Cobol programs, shell scripts, etc. that regularly update data in data warehouse. These tools are also helpful to maintain the Metadata.

These ETL Tools have to deal with challenges of Database & Data heterogeneity.

METADATA:

The name Meta Data suggests some high- level technological concept. However, it is quite simple. Metadata is data about data which defines the data warehouse. It is used for building, maintaining and managing the data warehouse.

In the Data Warehouse Architecture, meta-data plays an important role as it specifies the source, usage, values, and features of data warehouse data. It also defines how data can be changed and processed. It is closely connected to the data warehouse.

QUERY TOOLS:

One of the primary objects of data warehousing is to provide information to businesses to make strategic decisions. Query tools allow users to interact with the data warehouse system.

These tools fall into four different categories:

Query and reporting tools
Application Development tools
Data mining tools
OLAP tools

Characteristics of OLAP:

Multidimensional conceptual view: OLAP systems let business users have a dimensional and logical view of the data in the data warehouse. It helps in carrying slice and dice operations.

Multi-User Support: Since the OLAP techniques are shared, the OLAP operation should provide normal database operations, containing retrieval, update, adequacy control, integrity, and security.

Accessibility: OLAP acts as a mediator between data warehouses and front-end. The OLAP operations should be sitting between data sources (e.g., data warehouses) and an OLAP front-end.

Storing OLAP results: OLAP results are kept separate from data sources.

Uniform documenting performance: Increasing the number of dimensions or database size should not significantly degrade the reporting performance of the OLAP system.

OLAP provides for distinguishing between zero values and missing values so that aggregates are computed correctly.

OLAP system should ignore all missing values and compute correct aggregate values.

OLAP facilitate interactive query and complex analysis for the users.

OLAP allows users to drill down for greater details or roll up for aggregations of metrics along a single business dimension or across multiple dimension.

OLAP provides the ability to perform intricate calculations and comparisons.

OLAP presents results in a number of meaningful ways, including charts and graphs.

OLAP Types:

Three types of OLAP servers are:-

1. Relational OLAP (ROLAP)
2. Multidimensional OLAP (MOLAP)
3. Hybrid OLAP (HOLAP)

1. Relational OLAP (ROLAP):

Relational On-Line Analytical Processing (ROLAP) work mainly for the data that resides in a relational database, where the base data and dimension tables are stored as relational tables. ROLAP servers are placed between the relational back-end server and client front-end tools. ROLAP servers use RDBMS to store and manage warehouse data, and OLAP middleware to support missing pieces.

Advantages of ROLAP:

1. ROLAP can handle large amounts of data.
2. Can be used with data warehouse and OLTP systems.

Disadvantages of ROLAP:

1. Limited by SQL functionalities.
2. Hard to maintain aggregate tables.

2. Multidimensional OLAP (MOLAP):

Multidimensional On-Line Analytical Processing (MOLAP) support multidimensional views of data through array-based multidimensional storage engines. With multidimensional data stores, the storage utilization may be low if the data set is sparse.

Advantages of MOLAP

1. Optimal for slice and dice operations.
2. Performs better than ROLAP when data is dense.
3. Can perform complex calculations.

Disadvantages of MOLAP

1. Difficult to change dimension without re-aggregation.
2. MOLAP can handle limited amount of data.

3. Hybrid OLAP (HOLAP):

Hybrid On-Line Analytical Processing (HOLAP) is a combination of ROLAP and MOLAP. HOLAP provide greater scalability of ROLAP and the faster computation of MOLAP.

Advantages of HOLAP

1. HOLAP provide advantages of both MOLAP and ROLAP.
2. Provide fast access at all levels of aggregation.

Disadvantages of HOLAP

1. HOLAP architecture is very complex because it support both MOLAP and ROLAP servers.

What is Data Mining?

Data mining is finding useful patterns and insights from a large volume of datasets, typically through various techniques from statistics, Machine learning, and Artificial Intelligence. The main objective of data mining is to extract valuable knowledge from large amounts of data and transform it into a more easy-to-understand structure for further use.

Data mining involves several steps, including data cleaning, integration, selection, transformation, data mining, pattern evaluation, and knowledge representation. These steps are iterative and interactive, as the results of one step can influence the following steps and the entire data mining process.

Some standard techniques used in data mining include clustering, classification, association rule mining, regression analysis, and anomaly detection. These techniques can be applied to a wide range

of data types, such as structured data (e.g., databases), unstructured data (e.g., text), and semi-structured data (e.g., XML).

The insights gained from data mining can be used in various applications, such as business intelligence, fraud detection, healthcare, and social media analysis. Data mining is valuable for extracting practical knowledge and insights from large datasets.

There are multiple reasons to learn Data Mining, and the reason may vary from person to person. Some of the common reasons are:

1. **To gain insights from large data sets:** Data mining enables you to extract valuable insights from large datasets that may not be immediately available. By applying Data mining techniques, one can uncover new patterns, trends, and relationships that can help make more informed decisions
2. **To improve decision-making:** With the help of data mining techniques, one can make more informed decisions that are backed by data rather than intuition or guesswork. This can help you have better decisions about everything from product development to marketing to customer services
3. **To identify opportunities:** Data mining can help to identify new insights that may not have otherwise been considered. By analyzing data from a variety of sources, one can find the trends and patterns that can help capitalize on emerging opportunities
4. **To increase efficiency:** Data mining can help one identify inefficiencies and bottlenecks in one's business processes, allowing them to streamline operations and improve efficiency
5. **To stay competitive:** In today's data-driven world, companies that refrain from using data mining techniques risk falling behind their competitors. By learning data mining techniques, one can stay ahead of the competition and use data to their advantage

How Data Mining Works

The process of data mining is to extract information from a large volume of datasets. It is done in four phases:

1. **Data acquisition:** It is the process of collecting, filtering, and cleaning the data before it is added to the warehouse
2. **Data cleaning, preparation, and transformation:** It is the process in which data is cleaned, preprocessed, and transformed after adding it to the warehouse
3. **Data analysis, modeling, classification, and forecasting:** In this step, data is analyzed with the help of various models, and classification is done
4. **Final report:** In this step, the final report, insights, and analysis are finalised

The Data Mining Process

Data mining requires domain knowledge, technical skills, and creativity to extract essential insights from giant data sets. The steps and techniques used in the process can vary depending on the data and the specific problem being solved. There are multiple steps involved in the process of data mining:

1. **Data collection:** The first step in data mining is collecting relevant data from various sources. This can include data from databases, spreadsheets, websites, social media, sensors, and other

sources

2. **Data preprocessing:** When the data is collected, it must be preprocessed. This involves cleaning the data, removing duplicates, and dealing with missing data
3. **Data exploration:** The next step is to identify patterns, trends, and relationships. This can be done using various visualization techniques such as scatter plots, histograms, and box plots
4. **Data transformation:** In this step, the data is transformed into a format suitable for analysis. This can involve normalization, discretization, or other techniques
5. **Data modeling:** In this step, various algorithms and methods are used to build a model that can be used to extract insights from the data. This can include clustering, classification, regression, or other procedures
6. **Model evaluation:** Once the model is built, it must be evaluated to determine its efficiency. This involves testing the model on a subset of the data and comparing the results to known outcomes
7. **Deployment:** Finally, the model is deployed for a real-world problem, and the insights gained from the data mining process are used to make informed decisions

Types of Data Mining

Data mining can be broadly categorized into two types - predictive and descriptive data mining analysis.

Predictive Data Mining Analysis

Predictive data mining analysis is a sort of data mining that involves analyzing historical data to forecast future events or trends. It entails developing models that use various statistical and machine learning algorithms to estimate future events based on trends discovered in previous data. Predictive data mining models can be used to predict a variety of scenarios including anticipating customer behavior, detecting potential fraud, projecting revenue, and estimating the risk of a disease appearing in a specific group.

Predictive data mining frequently involves a significant amount of data preparation and preprocessing as well as the selection of relevant algorithms to develop models that can effectively predict future outcomes. Predictive data mining approaches can be further classified into the following categories:

- Analysis of Regression
- Analysis of classification
- Analysis of time series

Descriptive Data Mining Analysis

Descriptive data mining analysis is a sort of data mining that focuses on discovering and comprehending the underlying patterns and relationships in a dataset. Unlike predictive data mining, descriptive data mining is focused on summarizing and displaying data to acquire insights into its structure and properties rather than making predictions about future events.

Descriptive data mining techniques are frequently used for exploratory data analysis, discovering patterns and relationships in data and gaining a better understanding of the data's underlying distribution. Descriptive data mining approaches can be further classified into the following categories:

- Analysis of clusters

- Analysis of summarization and visualization
- Mining is governed by an association
- Mining for sequential patterns

Applications of Data Mining

Data mining has a wide range of applications across different industries. Here are some of the most common applications of data mining:

1. **Customer Relationship Management:** Data mining is used in customer relationship management (CRM) to analyze customer data and identify patterns and trends in customer behavior. This can help businesses better understand their customers and develop more effective marketing and sales strategies
2. **Fraud Detection:** Data mining is used in fraud detection to identify unusual patterns and anomalies in data that may indicate fraudulent activity. This can help businesses to detect and prevent fraudulent transactions, such as credit card fraud and identity theft
3. **Healthcare:** Data mining is used in healthcare to analyze patient data and identify patterns and trends in patient behavior and treatment outcomes. This can help healthcare providers to develop more effective treatment plans and improve patient outcomes
4. **Market Research:** Data mining is used in market research to analyze customer data and identify patterns and trends in customer behavior and preferences. This can help businesses to develop more effective marketing and sales strategies and stay ahead of the competition
5. **Predictive Analytics:** Data mining is used in predictive analytics to identify patterns and trends in data that can be used to make predictions about future events or outcomes. This can help businesses to make more informed decisions and anticipate future trends and developments
6. **Financial Analysis:** Data mining is used in financial analysis to analyze financial data and identify patterns and trends in financial markets and investment performance. This can help investors to make more informed investment decisions and manage risk more effectively
7. **Sports Analysis:** Data mining is used in sports analysis to analyze player and team data and identify patterns and trends in player performance and team dynamics. This can help coaches and managers to make more informed decisions and develop more effective game strategies

Association in Data Mining

Association in data mining identifies relationships between items in a dataset. This technique is commonly used in market basket analysis to identify which items are frequently purchased together. The output of association mining is a set of rules that indicate the strength of the association between items.

1. **Support and Confidence:** Support calculates the frequency with which an item set appears in the dataset. It is calculated as the ratio of transactions that contain a particular item to the total number of transactions in the dataset. While confidence measures the strength of a rule between two items. It is the ratio of the number of transactions that contain both the previous and consequent of the rule to the number of transactions that contain only the antecedent

2. **A-priori algorithm:** Apriori algorithm is used in data mining to discover frequent items in large datasets. It is an unsupervised learning algorithm that is commonly used for market basket analysis to identify items that are frequently purchased together. The algorithm works by generating a set of items and then pruning the item that does not meet the minimum support threshold
3. **FP Growth algorithm:** The FP-Growth algorithm is a popular algorithm used to discover frequent items in large datasets. It is an unsupervised learning algorithm that works by building a compact data structure called an FP-tree to represent the transactional database and then recursively mining the tree to generate frequent itemsets

Clustering in Data Mining

1. **Similarity and distance measure:** Similarity and distance measures are important concepts in data mining and are used to measure the similarity or dissimilarity between objects or data points. These measures are commonly used in clustering, classification, and recommendation systems. Basically, similarity measure is a method of calculating how the data samples are related to each other. A dissimilarity measure is used to figure out how much the data objects are distinct
2. **Partitioning methods:** Partitioning Methods are one of the most widely used clustering techniques in data mining. It involves dividing a dataset into several non-overlapping partitions, with each partition representing a cluster. There are several partitioning methods in data mining:
 1. **K-Means & Medoids:** The k-means algorithm partitions the dataset into k clusters, where k is a pre-defined number. The algorithm starts by randomly selecting k centroids and assigning each point in the dataset to the nearest centroid. The centroids are then recalculated as the mean of the points assigned to them, and the process is repeated until the centroids no longer change or the maximum number of iterations is reached. Medoids is another variation of the k-means algorithm, where instead of using the mean of the points, the representative point of each cluster is selected as the point that minimizes the average distance between it and all other points in the cluster
 2. **Hierarchical:** Hierarchical clustering involves creating a hierarchy of clusters by recursively dividing or merging clusters based on a similarity measure. The two main types of hierarchical clustering are agglomerative (bottom-up) and divisive (top-down). In agglomerative clustering, each data point initially forms its own cluster, and then pairs of clusters are successively merged based on a distance metric until a single cluster containing all the data points is obtained. In divisive clustering, all data points initially belong to a single cluster, and clusters are successively divided into smaller clusters based on a distance metric until each data point is in its own cluster
 3. **Density-Based:** Density-based clustering involves identifying regions of high density in the dataset and partitioning the data points into clusters based on these regions. Density-based clustering algorithms are particularly useful when the clusters are of arbitrary shape and size. The two most widely used density-based clustering algorithms are DBSCAN (Density-Based Spatial Clustering of Applications with Noise) and OPTICS (Ordering Points To Identify the Clustering Structure)

Classification Techniques

Classification is a data mining technique that involves identifying the class or category to which an object or observation belongs. It is a supervised learning method which uses a set of labeled training data to create a predictive model which can be used to classify new or unseen data. Features or attributes are the characteristics of the data that are used to determine its class or category. In classification, features are the variables or attributes that are used to predict the class label of the target variable. Some examples of features in data mining include:

1. **Numeric features:** These are numerical variables such as age, income, and temperature
2. **Categorical features:** These are variables that represent categories, such as gender, race, or type of product
3. **Binary features:** These are variables that take on one of two possible values, like yes or no, true or false, or 0 or 1
4. **Text features:** These are variables that represent text data, such as customer reviews or product descriptions
5. **Image features:** These are variables that represent visual data, such as pixel values in an image
6. **Time-series features:** These are variables that represent data collected over time, like stock prices or weather data
7. **Geospatial features:** These are variables that represent spatial data, such as latitude and longitude or elevation

Types of Classification

1. **Decision tree induction:** Decision tree induction is a popular classification technique in data mining that uses a tree-like model to represent decisions and their possible consequences. It is a supervised learning method that builds a decision tree from a set of training data to predict the class label of new data instances. Let's see an example:

Outlook	Temperature	Activity
Sunny	Hot	Don't play
Rainy	Cold	Don't play
Sunny	Mild	Play
Rainy	Mild	Play

The decision tree for this kind of data set would be:

2. Bayesian classification & Bayesian Belief Networks:

Bayesian Classification: Bayesian classification is a statistical method used to classify data into different classes based on the probability of each class. In this approach, the classifier computes the probability of each class given a set of input features, using Bayes' theorem. It assumes that the features are independent of each other and estimates the probabilities of each class based on the probability distribution of the features. Bayesian classification is commonly used in text classification, image recognition, and other applications. For example, in text classification, a Bayesian classifier might be trained to classify news articles into different categories, such as sports, politics, or entertainment, based on the words that appear in the article. The classifier would compute the probability of each category given the words in the article and assign the article to the category with the highest probability.

Bayesian Belief Networks(BBNs): They are graphical models that represent the relationships between variables and their probabilities. A BBN consists of a set of nodes, each representing a random variable, and directed edges connecting the nodes, representing the conditional dependencies between the variables. Each node has a conditional probability table (CPT) that specifies the probability of that node given the values of its parents. BBNs can be used for various tasks, such as classification, prediction, and decision-making under uncertainty. They are particularly useful when dealing with complex problems that involve multiple variables and uncertain relationships between them. For example, in medical diagnosis, a BBN could be constructed to represent the relationships between symptoms, diseases, and test results. The BBN would allow the system to reason about the likelihood of different diseases, given the observed symptoms and test results, and provide a diagnosis with a degree of certainty.

Lazy Learners: Lazy learners refer to machine learning algorithms that do not generalize the training data to construct a model. Instead, they memorize the training data and use it as a reference to make predictions on new instances. Lazy learners do not require the computational overhead of training models, and they can be faster than eager learners that construct a model based on the training data. One example of a lazy learner is the k-Nearest Neighbor (k-NN) algorithm. In k-NN, the algorithm simply stores the entire training dataset in memory and uses it to make predictions on new instances. When a new instance is presented to the algorithm, it finds the k closest instances in the training set based on some distance metric and predicts the class of the new instance based on the class labels of the k nearest neighbors.

Rule-based classification: Rule-based classification is a data mining technique that uses a set of if-then rules to classify new instances. Here's an example of how rule-based classification can be used in practice:

Suppose we have a dataset of patients with different medical conditions, and we want to develop a rule-based classifier to predict whether a patient has diabetes based on their symptoms. The dataset includes the following variables: age, sex, BMI, blood pressure, cholesterol level, and glucose level. To develop a rule-based classifier, we would first divide the dataset into a training set and a test set. We would use the training set to generate a set of if-then rules that predict whether a patient has diabetes based on their symptoms. Here's an example of a rule that could be generated: If glucose level ≥ 126 AND BMI ≥ 25 AND age ≥ 45 . Then the patient is classified as having diabetes. This rule states that if a patient's glucose level is equal to or greater than 126 mg/dL, their BMI is equal to or greater than 25, and their age is equal to or greater than 45, they will be classified as having diabetes. We would generate several such rules based on the training data and evaluate their performance on the test data. If the classifier performs well, we can use the rules to predict whether new patients have diabetes based on their symptoms.

Prediction Techniques

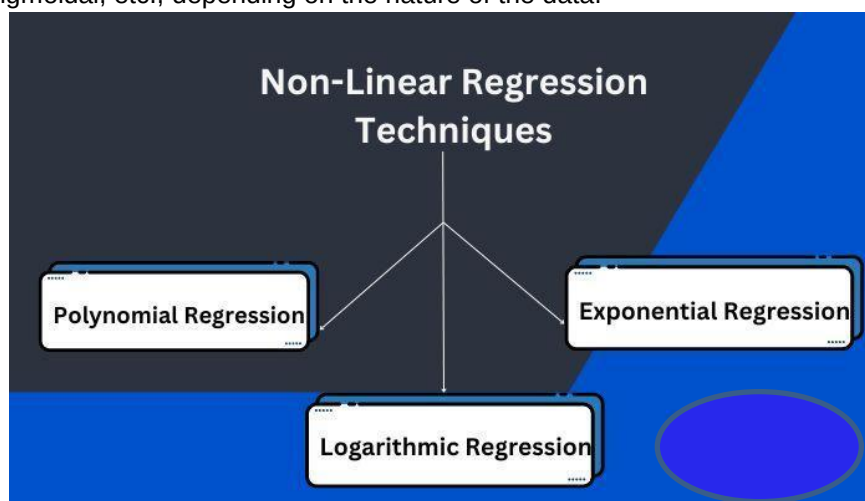
Predictive Data Mining is the analysis performed to forecast a future event, other data, or trends, as the name 'Predictive' means to forecast anything. Predictive Data Mining can help business analysts make better decisions and contribute to the analytics team's efforts. Predictive Data Mining assists in Predictive Analytics. As we all know, predictive analytics is the use of data to forecast outcomes.

Linear Regression Techniques

1. Linear regression is used to model the relationship between a dependent variable and one or more independent variables. Linear regression can be used for both supervised and unsupervised learning tasks
2. In supervised learning, linear regression is used for regression tasks where the goal is to predict a continuous numerical value. For example, predicting the price of a house based on its size, location, number of bedrooms, etc. In this case, linear regression can be used to find the best-fitting line or hyperplane that predicts the house price based on the given features
3. In unsupervised learning, linear regression is used for dimensionality reduction tasks. For example, in principal component analysis (PCA), linear regression is used to find the principal components that capture the most variance in the data. Linear regression can also be used for clustering tasks, where the goal is to group similar data points together based on their features
4. There are several techniques used to estimate the parameters of a linear regression model, including ordinary least squares (OLS), maximum likelihood estimation (MLE), and gradient descent. These techniques are used to find the best-fitting line or hyperplane that minimizes the sum of squared errors between the predicted values and the actual values of the dependent variable

Non-Linear Regression Techniques

Non-linear regression is a statistical technique used to model the relationship between a dependent variable and one or more independent variables when the relationship is non-linear. In non-linear regression, the model can take on various functional forms, such as quadratic, exponential, logarithmic, sigmoidal, etc., depending on the nature of the data.



Here are some examples of non-linear regression techniques:

1. **Polynomial Regression:** Polynomial regression is a non-linear regression technique that models the relationship between the dependent variable and the independent variable as an n th-degree polynomial function. It is used to capture complex, non-linear relationships between the variables. For example, a quadratic equation can be used to model the relationship between a

person's age and income

2. **Exponential Regression:** Exponential regression is a non-linear regression technique that models the relationship between the dependent variable and the independent variable as an exponential function. It is used to model growth or decay processes, such as population growth or radioactive decay. For example, the growth of a bacteria population over time can be modeled using an exponential function
3. **Logarithmic Regression:** Logarithmic regression is a non-linear regression technique that models the relationship between the dependent variable and the independent variable as a logarithmic function. It is used to model data that exhibits a logarithmic trend, such as the relationship between the number of years of education and a person's income

Advanced Mining Applications

Mining Data Streams

Data stream mining is the process of analyzing data that arrives in a continuous and rapid stream. For example, social media updates, sensor data, or network traffic. It involves real-time analysis of data as it is generated, using algorithms that can handle high-volume, high-velocity, and high-variety data. It includes online learning, incremental clustering, sliding window analysis, and sketch-based algorithms. Machine learning techniques, such as decision trees, neural networks, and support vector machines, can also be used for data stream mining.

Applications of data stream mining include real-time monitoring of social media sentiment, predicting traffic congestion, detecting network intrusions, and optimizing online advertising. However, data stream mining poses challenges related to the high volume and velocity of data, the need for fast and efficient algorithms, and the potential for concept drift and data imbalance.

Mining Time Streams

Time series mining is the process of analyzing data that is collected over time to identify patterns, trends, and relationships. It is used in many different fields, including finance, healthcare, meteorology, and manufacturing, to make predictions, improve decision-making, and optimize operations. It includes statistical methods, such as time series analysis, autoregressive integrated moving average, and exponential smoothing, as well as machine learning methods, such as artificial neural networks, decision trees, and support vector machines.

Applications of time series mining include forecasting stock prices, predicting patient outcomes in healthcare, detecting anomalies in sensor data, and optimizing manufacturing processes. However, time series mining also poses challenges related to the complexity and variability of time series data, the need for domain-specific knowledge, and the potential for overfitting and bias.

Web Mining

Web mining is the process of extracting valuable information from the World Wide Web by analyzing the content of web pages, the structure of the web, and the usage of the web. It involves data mining techniques to identify patterns, trends, and relationships within large datasets of web data. It has numerous applications in different fields, such as e-commerce, marketing, social media analysis, and personalized recommendations. It enables organizations to understand user behavior, improve customer experience, and gain insights into market trends. However, web mining also raises concerns about privacy and security and ethical issues related to the collection and use of user data.

Also read anomalies in database

Examples of Data Mining

Some of the common examples of Data mining are:

1. **Retail:** Retailers use data mining to analyze customer transactions to discover patterns and trends that can be used to improve customer loyalty and increase sales. For example, a retailer might use data mining to discover that customers who buy a specific product also tend to buy another product, allowing the retailer to make recommendations to customers based on their buying habits
2. **Healthcare:** Healthcare providers use data mining to analyze patient data to discover patterns and relationships that can help diagnose diseases and build treatment plans. For example, a healthcare provider might use data mining to find that patients with specific symptoms are more likely to develop a particular disease, allowing the provider to create a screening program to catch the disease early
3. **Finance:** Financial institutions utilize data mining to analyze customer data to discover patterns and trends that can be used to detect fraud and improve risk management. For example, a financial institution might use data mining to learn that customers who make large cash withdrawals are more likely to be victims of fraud, allowing the institution to implement additional security measures to protect these customers

Advantages of Data Mining

1. **Predictive analysis:** Data mining allows businesses to predict future trends and behaviors based on historical data. This enables organizations to make better decisions about future strategies, products, and services
2. **Improved marketing:** Data mining helps businesses identify customer behavior and preference patterns. This can help organizations create targeted marketing campaigns and personalized offers that are more likely to resonate with customers
3. **Improved customer experience:** Data mining can help businesses understand customer preferences and behaviors, enabling organizations to tailor products and services to meet their needs. This can result in higher customer satisfaction and loyalty
4. **Competitive advantage:** Data mining enables businesses to gain insights into their competitors' strategies and performance. This can help organizations identify areas where they can earn a competitive advantage and differentiate themselves in the marketplace
5. **Increased efficiency:** Data mining can help businesses streamline processes and operations by identifying inefficiencies and bottlenecks. This can help organizations optimize workflows and reduce costs
6. **Fraud detection:** Data mining can help detect fraudulent activities and patterns in financial transactions. This can help organizations prevent financial losses and maintain the integrity of their operations

Disadvantages of Data Mining

1. **Cost:** Data mining can be expensive, as it requires significant computing power and resources to analyze large datasets. Small businesses or organizations with limited budgets may need help implementing data mining
2. **Complexity:** Data mining is a complex process that requires specialized knowledge and expertise. Analyzing large and complex datasets can be challenging, and it may require a team

of data scientists to develop and implement effective data mining techniques

3. **Bias and inaccuracies:** Data mining can uncover biased or discriminatory patterns if the data used in the analysis is biased or incomplete. This can lead to incorrect or unfair conclusions and decisions
4. **Ethical concerns:** Data mining raises ethical concerns about the use of personal data and privacy issues. Organizations must be transparent about their data collection and use practices and ensure that they comply with relevant laws and regulations
5. **Over-reliance on technology:** Data mining can lead to an over-reliance on technology, which may result in a lack of human judgment and intuition. Human interpretation and analysis are essential to ensure that data mining results are accurate and meaningful
6. **Data quality:** Data mining requires high-quality data to produce reliable and valuable results. If the data is incomplete, inconsistent, or of poor quality, the data mining results may not be accurate or valid

Data Mining vs. Data Analytics and Data Warehousing

	Data Mining	Data Warehousing
Key Functions	Pattern Recognition, Anomaly Detection and Predictive Analysis	Data Integration, Data Storage, and Data Retrieval
Techniques	Classification, Clustering, Regression, Association and Rule Learning	ETL (Extract, Transform, Load), OLAP (Online Analytical Processing), and Data Modeling
Popular Tools used	R, Python	Snowflake, Microsoft SQL Server
Focus and	Focuses on patterns and insights within	Creates a

Scope	the data	centralized, organized, accessible repositor	and data
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What Are Association Rules in Data Mining?

Defined by their names, association rules are if/then statements that identify the relationships or dependencies between the data. With the characteristic property of suiting numeric and non-numeric categorical data, it is often applied in market basket analysis and other applications. It can uptake data from relational and transactional databases and other data sources.

The association rule has two parts: antecedent or if and consequent or then. The antecedent is the first part available in data, while the resulting is the resultant part available in combination with the antecedent. For instance, the example of market basket analysis will be: "If a customer buys running shoes, then there is a likelihood that they will also buy Energy bars." here, running shoes are antecedent, and energy bars are consequent. The example more particularly targets the fitness enthusiast audience.

What Are Use Cases for Association Rules in data Mining?

There is a wide variety of applications for association rules. The top three association rules in data mining examples are:

Market Basket Analysis: An example of a shopping combination can be a purchase of yogurt, and granola is likely to be associated with purchasing berries. It indicates the significance of the association rule in analyzing purchasing habits and requirements. The practical usage of interpretation is seen in developing suitable combination offers, optimizing product placements, and increasing sales.

Fraud Detection: Here, the combination of use is identifying a purchase pattern, its location, and frequency. Recognition of the same helps flag fraudulent activities and take preventive measures from the same IP address.

Recommendation systems: These include detecting the usage patterns from browsing history and previous purchases to predict the future requirements of the user. The recommendations are based on the same. Expanding the usage from marketing is significant in music and shows-based services as well.

The prediction in the association rule explained previously with examples, is calculated based on cardinality, support, and confidence. Cardinality refers to the relation between two items, which proportionally increases with the number of objects. The support indicates the frequency of the statements, and then the confidence informs the frequency of truthfulness of these relationships. Explain the association rules work by determining the rules governing the reason and situation where the combination may occur. For instance, the preferred healthy and less time-consuming breakfast option combines yogurt with granola and berries.

Often, in practical situations, the numbers get unrealistic. Some statistically independent items with the least purchase combination might come together with a significantly high percentage in practical usage. For instance, statistically, lesser chances of combined purchase of beer and diapers occur while real-world statistics are comparatively higher. The increase in statistics is a

Association Rule Algorithms

Three algorithms generate association rules. These are stated as follows:

Apriori Algorithm

The association rules in the apriori algorithm are generated through frequent transaction datasets. Often used for market basket analysis, it uses techniques like Breadth-first search and Hash tree. Providing the information on combined products bought together, it also serves medical purposes by finding drug reactions for patients.

Eclat Algorithm

Also known as Equivalent Class Transformation, it uses a depth-first search technique. Providing quick and accurate execution, it also deals with transaction databases. The ELCAT algorithm uses less storage and works without repeated scanning of data for computing the individual support values. Instead, it uses transaction ID Sets or Tidsets for computation purposes.

F-P Growth Algorithm

Referred to as Frequent pattern growth, it is a further enhanced version of the Apriori algorithm. It is analyzed through two steps. The first is database conversion into a tree structure, thus earning the name due to the depiction of frequent patterns. The second step is the representation format, which further eases extracting the most frequent patterns.

Information Retrieval System

An Information Retrieval System is a set of algorithms that facilitate the relevance of displayed documents to searched queries. In simple words, it works to sort and rank documents based on the queries of a user. There is uniformity with respect to the query and

Types of Information Retrieval Model

There are several information retrieval techniques and types that can help you with the process. An information retrieval comprises of the following four key elements:

1. **D** – Document Representation.
2. **Q** – Query Representation.
3. **F** – A framework to match and establish a relationship between D and Q.
4. **R** (q, d_i) – A ranking function that determines the similarity between the query and the document to display relevant information.

There are three types of Information Retrieval (IR) models:

1. Classical IR Model — It is designed upon basic mathematical concepts and is the most widely-used of IR models. Classic Information Retrieval models can be implemented with ease. Its examples include Vector-space, Boolean and Probabilistic IR models. In this system, the retrieval of information depends on documents containing the defined set of queries. There is no ranking or grading of any kind. The different classical IR models take Document Representation, Query representation, and Retrieval/Matching function into account in their modelling. This is one of the most used Information retrieval models.

2. Non-Classical IR Model — They differ from classic models in that they are built upon propositional logic. Examples of non-classical IR models include Information Logic, Situation Theory, and Interaction models. It is one of the types of information retrieval systems that is diametrically opposite to the conventional IR model.

3. Alternative IR Model — These take principles of classical IR model and enhance upon to create more functional models like the Cluster model, Alternative Set-Theoretic Models Fuzzy Set model, Latent Semantic Indexing (LSI) model, Alternative Algebraic Models Generalized Vector Space Model, etc.

Let's understand the most-adopted similarity-based classical IR models in further detail:

1. Boolean Model — This model required information to be translated into a Boolean expression and Boolean queries. The latter is used to determine the information needed to be able to provide the right match when the Boolean expression is found to be true. It uses Boolean operations AND, OR, NOT to create a combination of multiple terms based on what the user asks. This is one of the information retrieval models that is widely used.

2. Vector Space Model — This model takes documents and queries denoted as vectors and retrieves documents depending on how similar they are. This can result in two types of vectors which are then used to rank search results either

- Binary in Boolean VSM.
- Weighted in Non-binary VSM.

3. Probability Distribution Model — In this model, the documents are considered as distributions of terms and queries are matched based on the similarity of these representations. This is made possible using entropy or by computing the probable utility of the document. They are of two types:

- Similarity-based Probability Distribution Model
- Expected-utility-based Probability Distribution Model

4. Probabilistic Models — The probabilistic model is rather simple and takes the probability ranking to display results. To put it simply, documents are ranked based on the probability of their relevance to a searched query. This is one of the most basic information retrieval techniques used.

Components of Information Retrieval Model

Here are the prerequisites for an IR model:

1. An automated or manually-operated indexing system used to index and search techniques and procedures.
2. A collection of documents in any one of the following formats: text, image or multimedia.
3. A set of queries that serve as the input to a system, via a human or machine.
4. An evaluation metric to measure or evaluate a system's effectiveness (for instance, precision and recall). For instance, to ensure how useful the information displayed to the user is.

If you draw and explain the IR system block diagram, you will come across different components. The various components of an Information Retrieval Model include:

Step 1

Acquisition

The IR system sources documents and multimedia information from a variety of web resources. This data is compiled by web crawlers and is sent to database storage systems.

Step 2

Representation

The free-text terms are indexed, and the vocabulary is sorted, both using automated or manual procedures. For instance, a document abstract will contain a summary, meta description, bibliography, and details of the authors or co-authors. It is one of the components of the information retrieval system that involves summarizing and abstracting.

Step 3

File Organization

File organization is carried out in one of two methods, sequential or inverted. Sequential file organization involves data contained in the document. The Inverted file comprises a list of records, in a term by term manner. It is one of the components of information retrieval system that also involves the combination of the sequential and inverted methods.

Step 4

Query

An IR system is initiated on entering a query. User queries can either be formal or informal statements highlighting what information is required. In IR systems, a query is not indicative of a single object in the database system. It could refer to several objects whichever match the query. However, their degrees of relevance may vary.

- Information Retrieval System Examples
- Search Engines: Google, Bing, and Yahoo
- Digital Libraries: PubMed, IEEE Xplore
- Database Management Systems: SQL databases like MySQL, Oracle
- Enterprise Search Systems: SharePoint, Elasticsearch
- E-commerce Search: Amazon's product search, eBay's search engine

Importance of Information Retrieval System

- **Productive and Efficient** – It is unproductive and possibly expensive for small businesses and local companies to have an owner or employee spend time looking through piles of loose papers or attempting to find records that are missing or have been improperly filed. In addition to lowering the likelihood of information being misfiled, robust information storage and retrieval system that includes a strong indexing system also accelerates the storing and information extraction. This time-saving advantage results in increased office productivity and efficiency while lowering anxiety and stress.
- **Regulatory Compliance** – A privately owned corporation is exempt from the majority of federal and state compliance regulations, unlike a public company. Despite this, many people decide to voluntarily comply in order to increase accountability and the company's reputation in public. Additionally, small-business owners are required to retain and maintain tax information so that it is easily available in the event of an audit. A well-organized system for information retrieval in Artificial Intelligence that adheres to compliance rules and tax record-keeping requirements greatly boosts a business owner's confidence that the operation is entirely legal.
- **Manual vs. Electronic** – The value of electronic information retrieval in Artificial Intelligence is based on the fact that they demand less storage space and cost less in terms of both equipment and manpower. An ordered file system may be maintained using a manual approach, but it requires financial allotments for storage space, filing equipment, and administrative costs. Additionally, an electronic system may make it much simpler to implement and maintain internal controls intended to prevent fraud, as well as make sure the company is adhering to privacy regulations.
- **Better Working Environment** – Anyone passing through an office space may find it depressing to see important records and other material piled on top of file cabinets or in boxes close to desks. Not only does this lead to a tense and unsatisfactory work atmosphere, but if consumers witness this, it could give them a bad impression of the company. To understand how crucial it is for even a small firm to have efficient information storage and retrieval system.

Difference Between Information Retrieval and Data Retrieval

Parameter	Information Retrieval	Data Retrieval
Retrieval Method	Based on the similarity between query and document	Based on keywords in user-entered query
Tolerance for Errors	Minor errors are tolerated and may go unnoticed	There is no room for errors; which leads to system failure
Structure	Ambiguous with no defined structure	Has a defined structure based on semantics

User Solutions	Does not provide solutions to user	Provides solutions to user
Result Precision	Produces approximate results	Produces exact results
Result Sorting	Displayed results sorted by relevance	Displayed results not sorted by relevance
Nature of Model	Probabilistic	Deterministic

Understanding the Difference between IRS and DBMS

Category	DBMS	IRS
Data Modelling Facility	A DBMS comes with an advanced Data Modeling Facility (DMF) that offers Data Definition Language and Data Manipulation Language.	The Data Modeling Facility is missing in an information retrieval system. In an IRS, data modeling is limited to the classification of objects.
Data Integrity Constraints	The Data Definition Language of DBMS can easily define the data integrity constraints.	These validation mechanisms are less developed in an information retrieval system.
Semantics	A DBMS offers precise semantics.	The semantics offered by an information retrieval system is usually imprecise.
Data Format	A DBMS comes with a structured data format.	An information retrieval system will have an unstructured data format.
Query Language	The query language of a DBMS is artificial.	The query language of an information retrieval system is extremely close to natural language.
Query Specification	In a DBMS, query specification is always complete.	Query specification is incomplete in an IRS.

Pros of Information Retrieval System:

- **Fast Answers:** Retrieval systems are like speed demons. They can zip through massive piles of data and fetch you the info you need in a flash.
- **Organized Chaos:** They turn data chaos into order. Imagine having a messy room – retrieval systems tidy up your digital space, making it easier to find stuff.
- **24/7 Availability:** These systems don't take vacations. They're always on, ready to pull up information whenever you need it, day or night.
- **Tailored Results:** They're like personal shoppers for data. Based on your needs, retrieval systems bring you results that match your requirements, saving you from info overload.
- **Collaboration Boost:** Retrieval systems make sharing a breeze. Whether it's documents or data, they help teams collaborate by ensuring everyone has access to the right info.

Cons of Information Retrieval System:

- **Garbage In, Garbage Out:** If the data going in is a mess, the results will be too. Retrieval systems heavily rely on the quality of input data.
- **Overreliance on Keywords:** Sometimes, these systems can be a bit like keyword junkies. If your search terms don't match exactly, you might miss out on crucial information.
- **Privacy Concerns:** With great power comes great responsibility. Retrieval systems handling sensitive data can raise privacy concerns, requiring robust security measures.
- **Learning Curve:** They can be a bit like a new gadget – takes a while to figure out. Users might need some time to get the hang of using these systems effectively.
- **Information Overload Risk:** While they help organize, retrieval systems can also overload you with too much info. It's like having a well-organized closet but not remembering where you put that one shirt.
- **Maintenance Headaches:** Just like a car needs regular checkups, retrieval systems need maintenance. Ensuring they stay efficient and updated can be an ongoing task.